Residual $SO(4,2)$ Cohomology of Free Weyl Gravity on the Conformal Cylinder: Cartan Homotopy, Weyl-Square Classes, and Their Residual Pairing

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Abstract

We compute the residual cohomology of free four-dimensional Weyl gravity on the conformal cylinder $\mathbb{R} \times S^3$ in a selected closed-universe problem that treats all fifteen conformal transformations—including compact time translation D—as residual gauge transformations. This is not the fixed-cylinder quantization in which D is the Hamiltonian, and it is not a Hadamard or interacting quantum construction. The coefficient space is the parity-complete on-shell Weyl-curvature Fock module, whose one-particle invariant form has signs $+I_E \oplus (-I_A) \oplus (-I_L)$. Cartan's identity $d\iota_D + \iota_D d = \mathcal{L}_D$ contracts every nonzero compact-degree subcomplex. In the centered four-ghost polarization, the vacuum and one-particle sectors are acyclic and the complete two-particle calculation gives

$$H^4_{\mathrm{res}} = \mathrm{span}\{[W^2_+], [W^2_-]\}, \qquad \mathrm{sig}\, G_{\mathrm{res}} = (2,0), \qquad G_{\mathrm{res}} = I_2.$$

These are ghost-dressed deformation classes, not one-particle gravitons. Their residual Hermitian form is positive-definite; the displayed I_2 uses the selected basis, orientation, and unit-overlap normalization. Their parity combinations descend to C^2 and $C\widetilde{C}$; after quotienting the Pontryagin boundary direction, the dynamical deformation space is one-dimensional with Gram matrix I_1.

We connect this calculation to the covariant free BV complex. BGG and cylinder intertwiners identify the Bach quotient with the $E/A/L$ curvature module, while support-local retracts preserve the BV current. Null symbol ranks 11 and 9 rule out a scalar-wave witness on the original field bundle and isolate helicity ± 2. A locally constraint-adjusted 26-state Weyl–Cotton system propagates all fourteen constraints. Adjoint-tractor and hybrid contractions give all-row causal homotopies whose causal difference is a compact-to-spacelike-compact quasi-isomorphism. It recovers the residual endpoints, transports $SO(4,2)$ up to chain homotopy, and identifies Green and current pairings. Hence $H^4_{\mathrm{cov}} = \mathrm{span}\{[W^2_+], [W^2_-]\}$ and $\mathrm{sig}\, G_{\mathrm{cov}} = (2,0)$, with $G_{\mathrm{cov}} = I_2$ after the same normalization choices.

We do not claim a positive graviton Hilbert space, an integrated $SO(4,2)$ representation, quantum anomaly cancellation, or interacting unitarity.

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Authorship and generated inputs

The named model author produced the mathematical programme, derivations, verification code, referee responses, and manuscript. Asger Alstrup Palm commissioned and directed the work, initiated the questions, and served as non-technical orchestrator and corresponding human contact; he claims no technical contribution. Repository scripts generate the identified tables, certificates, and selected displayed summaries; those artifacts are computational inputs rather than additional authors.

Contents

1 Introduction

Pure conformal gravity in four dimensions is defined, up to an overall coupling and the Euler density, by the Weyl-square action

$$S_W[g] = -\alpha_g \int_M d^4x \sqrt{-g}\, C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma}. \tag{1}$$

Its fourth-order linearized equation has more solutions than the Einstein equation, and a conventional local reduction produces an indefinite oscillator form. This familiar observation does not by itself determine the state space after all gauge constraints have been imposed. On a compact spatial slice, background conformal symmetries are reducibilities of the local Diff×Weyl gauge system. Treating those residual transformations as gauge introduces a second, finite-dimensional reduction which is absent from a naive oscillator count. It is also a substantive physical choice. In a Dirac or background-independence interpretation of a closed universe, all residual conformal transformations can be quotiented. In a fixed cylinder conformal field theory, by contrast, D is ordinarily the Hamiltonian and the conformal group acts globally. This paper computes the first choice; it does not assert that strict pure-Weyl gravity uniquely selects it.

The organizing question is therefore more basic than a sign assignment: before asking whether a gravitational mode has positive or negative norm, one must first determine whether it survives the full gauge reduction being imposed. The answer below concerns the residual reduction. The smooth metric-to-curvature bridge is proved globally, and a symbolic all-energy cylinder construction identifies its algebraic positive-energy part with the $E/A/L$ module used below. The passage to the complete selected algebraic local-plus-residual BV–BFV state complex is also proved below. Its natural energy-mode Krein–Fock completion is proved as well. The reduced Lorentzian metric fields are then identified with that completion by a field-induced branch Cauchy–Sobolev theorem. The auxiliary route now supplies the curved four-row operator identities, a support-preserving all-row deformation retract, and the covariant current comparison. It also supplies a basis-independent scalar-wave no-go theorem: at a null covector the action Hessian has rank 11, whereas the gauge symbol has rank 9. The uncancelled rank-two quotient is precisely helicity ± 2, and the reduced Weyl symbol acts on it by $\frac{1}{4}I_2$. Thus a causal contraction must carry the physical sector through curvature or tractor variables rather than hide it in a 24-component scalar-wave block. The Weyl–Cotton equations, their constraint-adjusted symmetric-hyperbolic closure, sourced constraint propagation, and the all-level $E/A/L$ audit are proved. The support-local all-row prolongation retract and prolonged BV differential are coefficientwise complete as well. The curved adjoint-tractor BGG retract transfers the parent causal homotopy to the trace-free endpoint; a local trace/Weyl shear and the hybrid algebraic projector extend it to all 386 prolonged rows. The induced causal map is now certified as a quasi-isomorphism, all fifteen residual endpoints and their duals are recovered without duplication, the $SO(4,2)$ action transfers up to an explicit compactly supported chain homotopy, and the direct cyclic Green/current comparison fixes the covariant Gram matrix. Thus the covariant H^4 theorem is transport of the existing residual calculation, not a second cohomology computation. Distributional/Hadamard completion and alternative boundary polarizations remain open. A canonical endpoint Green inverse is not part of the transferred construction, while a direct invariant two-wave metric factorization is excluded in the certified family.

This paper starts from the classical linearized BV–BFV complex and then, after the stated positive-frequency polarization, studies its completed free BRST state representation on

$$M = \mathbb{R} \times S^3, \qquad \bar{g} = -dt^2 + d\Omega_3^2, \tag{2}$$

with unit cylinder radius. Three spaces must be distinguished:

$$\begin{aligned} \text{local oscillator cohomology} &\supseteq \text{classically integrable perturbations}, \\ \text{classically integrable perturbations} &\longrightarrow \text{residual-BRST classes}. \end{aligned} \tag{3}$$

The first contains the familiar Einstein, vector-descendant, and logarithmic or higher-derivative towers. The last is obtained only after the residual $SO(4,2)$ constraints and their ghosts are included. A linearized oscillator need not survive this final reduction.

Our result is most naturally stated in the language of residual vertex cohomology. The Hamada ghost polarization supplies a top-degree four-ghost factor of compact degree -4. It pairs with scalar matter primaries of weight four. The complete centered residual cohomology consists of two chiral Weyl-square classes, and their residual CE Hermitian form is positive-definite and normalized to the identity matrix I_2. Their parity combinations are the dynamical Weyl-square density and the topological Pontryagin density. Thus the positive matrix is attached to classical vertex or theory-space directions. It is not a positive norm on a propagating graviton Fock space; indeed, the one-particle residual cohomology vanishes.

Relation to previous work. The conformal deformation detour complex on obstruction-flat backgrounds was constructed by Branson and Gover [1]; its formal self-adjointness and related Yang–Mills detour constructions were developed in [3]. Gover and Peterson give the oriented four-dimensional deformation sequence, while Čap proves that the flat BGG sequence is a fine resolution and computes the same cohomology as the flat twisted de Rham complex [2, 7]. Cylinder harmonics, oscillator charges, and primary building blocks were developed by Hamada and Horata and by Hamada [10, 11]. In the broader conformal-mode/Riegert programme, Hamada identified the residual diffeomorphisms with cylinder conformal transformations [12], constructed the fifteen-generator residual BRST charge, used the product of four proper-conformal ghosts as the centered ghost vacuum, derived the scalar weight-four condition, and displayed a Weyl-square BRST state [13]. None of those ingredients is claimed as new here. We use the harmonic and residual-algebra conventions, but we do not import the Riegert field or the quantum claims of that model. The Weyl-curvature module and its character are consistent with the general conformal higher-spin description and partition-function literature [14, 15]. Finally, the identification of C^2 as the unique local four-derivative self-interaction of one linear conformal graviton, under the assumptions stated there, is due to Boulanger and Henneaux [18].

The defensible new result is narrower and more concrete. Starting with the parity-complete pure-Weyl coefficient module, we compute the complete minimal *absolute* residual complex in the centered degree. We prove that the vacuum and one-particle sectors vanish, that precisely two two-particle classes survive after the incoming-image test, and that their normalized Hermitian residual CE Gram matrix is I_2. The standard Cartan homotopy makes the particle-number inventory exhaustive, and the Euler–Lagrange map splits the two positive directions into one dynamical and one topological class.

Claim boundary. Eight statements must remain separate.

1. The *algebraic residual theorem* is exact for the minimal absolute $SO(4,2)$ Chevalley–Eilenberg complex built on the on-shell Weyl module, the stated centered ghost polarization, and full residual conformal gauging.
2. The smooth complexified metric-to-geometric-curvature bridge is an exact global theorem. Its restriction to the D-finite cylinder category is also exact: the explicit $E/A/L$ harmonics give same-energy metric preimages, nonzero curvature intertwiners, and character exhaustion. These statements concern the geometric and algebraic realizations; the energy-mode completion is a separate theorem below.
3. In the finite polynomial conformal category, an exact metric-BV calculation now includes the ghost and antifield detour rows, constructs a homotopy-equivariant retract, transfers the strict residual action, and reproduces the centered kernel and image. Cross-energy recursion constructs its invariant physical form in the raw basis; the raw cyclic retract and dressed isometry are exact through the centered window. The minimal rows are derived from the field master action, and a specified gauge-fixed nonminimal extension contracts back to them exactly. For the selected algebraic BV–BFV suspension and positive-frequency polarization, the endpoint map has scalar $+1$, the polarized row ledger is exhaustive, and the matter-times-ghost pairing transfers exactly. This is not a statement about a unique boundary polarization.
4. The normalized $E/A/L$ module has a canonical infinite-index energy-mode Krein completion, its bosonic Fock completion is Krein, and the residual BRST operator has a closed maximal block realization. The bounded Cartan contraction reduces its cohomology to the unchanged finite centered block. This mode-space theorem alone does not construct the reduced field Cauchy realization below or integrate the unbounded Lie-algebra action to $SO(4,2)$.

5. On the Lorentzian cylinder the tensor- and vector-curl identities give local normally hyperbolic factors for the reduced TT and transverse-vector operators. The lower TT, vector, and upper TT branches are exactly the $E/A/L$ towers. Their field-action residues determine a branch Cauchy–Sobolev space Krein-unitarily equivalent to the completed one-particle module. This reduced theorem is not yet a Green's witness for the complete BV complex.

6. The action-derived curved auxiliary Hessian and four-row operator identities, the local BV-canonical deformation retract, and the covariant current comparison are exact. They are recorded respectively by `curved_operator_identity`, `curved_deformation_retract`, and `curved_current_comparison`, all of which are true. The same calculation proves that the former scalar-wave witness ansatz is impossible; it does not supply causal Green homotopies.

7. The null-symbol quotient is the two-dimensional helicity-± 2 module and the reduced Weyl symbol is $\frac{1}{4}I_2$. The 26-state Weyl–Cotton equations, constraint-adjusted symmetric hyperbolicity, sourced constraint propagation and all-level $E/A/L$ spectrum are certified. The support-local all-row prolongation and BV operator identity are also certified. The retained 30-row metric endpoint and its canonical backward witness are coefficient-complete. A normalization audit gives scalar trace-free biwave symbol and reduces the open primal channel to a rank-nine operator. Its complete invariant two-wave-factor family is ruled out by an order-two polynomial certificate. The canonical upper curvature lift is coefficientwise complete, the canonical middle graph lift is obstructed, and the minimal cyclic relative saddle has zero Schur correction. A flat adjoint-tractor HPL compression extends to a curved cyclic BGG retract: ten invariant curvature channels cancel the former 48 entries, and the compressed parent middle operator equals the complete Bach block with normalization -2 at every derivative order. Parent causal transfer, an explicit trace/Weyl shear, and the 356/30 hybrid retract give a causal homotopy on all 386 rows. Thirteen curvature/current/causal/transport flags are true. The two remaining false flags concern only the superseded canonical endpoint Green inverse and prolonged-witness implementation; the direct tractor route does not require them. The exact dependency DAG therefore promotes `final_covariant_H4`.

8. Calling the resulting classes physical states additionally requires a boundary and gauge principle which makes all fifteen conformal generators, including D, gauge rather than global charges. We call them *candidate physical classes under full residual conformal gauging.*

No statement in this paper establishes quantum BRST nilpotency, anomaly cancellation, interacting pseudo-unitarity, or a positive asymptotic particle space.

The local-to-residual bridge belongs to the general subject of local BRST cohomology with antifields, gauge fixing, and descent [19, 21]. Metsaev's ordinary-derivative BRST–BV construction gives a concrete realization of conformal fields and their conformal algebra on fields and antifields [26]. Dneprov and Grigoriev construct a minimal nonlinear Weyl-jet model and compatible presymplectic structure [27]. Their curvature-dependent ghost transformations make that model a valuable comparison, but not an immediate proof that the *free* transferred residual differential is strict. Neither work supplies the global cylinder zero-mode normalization used here.

Selected boundary and gauging input

$\Sigma = S^3$ closed; zero boundary/asymptotic charge; no external clock or deparametrization; all fifteen residual reducibilities gauged; centered positive-frequency polarization.

Paper A spine: residual calculation

$E/A/L$ Weyl module with its invariant Krein form

$\Downarrow$

polarized free symmetric algebra and absolute residual CE complex

$\Downarrow$

Cartan contraction and exhaustive centered $N = 0, 1, 2$ calculation

$\Downarrow$

$H^4_{\mathrm{res}} = \langle [W_+^2], [W_-^2] \rangle$,
$\operatorname{sig} G_{\mathrm{res}} = (2, 0)$, $G_{\mathrm{res}} = I_2$ after normalization

$\Downarrow$

descent to $C^2, C\widetilde{C}$ and the one-dimensional dynamical quotient

Paper B spine: covariant transport

metric BV detour complex and global BGG bridge

$\Downarrow$

support-local auxiliary/curvature retracts and current comparison

$\Downarrow$

Weyl–Cotton evolution, sourced constraints, and all-row causal homotopy

$\Downarrow$

causal–Cauchy quasi-isomorphism, endpoint recovery, and $SO(4, 2)$-equivariant pairing transport

$\Downarrow$

$H^4_{\mathrm{cov}} \cong H^4_{\mathrm{res}}$ with the same signature and normalized Gram form

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The right spine imports the already computed residual module and pairing from the left spine; it does not repeat the Chevalley–Eilenberg calculation. Both conclusions remain classical, free, and dependent on the selected closed-universe boundary/gauging input above.

Figure 1: Dependency diagram separating the residual computation from its covariant causal transport.

2 The free local pure-Weyl complex

2.1 Gauge, curvature, and equation maps

Let $h_{\mu\nu}$ be a metric perturbation and let (ξ^μ, σ) be an infinitesimal Diff×Weyl parameter. The full linear gauge map is

$$K(\xi,\sigma)_{\mu\nu} = \mathcal{L}_\xi \bar{g}_{\mu\nu} + 2\sigma\bar{g}_{\mu\nu} = 2\bar{\nabla}_{(\mu}\xi_{\nu)} + 2\sigma\bar{g}_{\mu\nu}. \tag{4}$$

After quotienting the trace, it becomes the conformal Killing operator

$$(K_0\xi)_{\mu\nu} = 2\bar{\nabla}_{(\mu}\xi_{\nu)} - \frac{1}{2}\bar{g}_{\mu\nu}\bar{\nabla}\cdot\xi. \tag{5}$$

Let

$$C_1 h = \frac{d}{d\varepsilon} C[\bar{g} + \varepsilon h]\bigg|_{\varepsilon=0} \tag{6}$$

be the linearized Weyl-curvature map, and let B_{lin} denote the action-normalized linearized Bach operator. Formal adjoints use the spacetime action pairing on a domain for which temporal and spatial boundary terms vanish.

The conformal cylinder is conformally flat. Naturality and conformal covariance therefore give the exact local identity

$$\boxed{C_1 K = 0} \tag{7}$$

for arbitrary local gauge parameters, not only for the fifteen residual conformal Killing parameters.

Proposition 2.1 (Action-normalized factorization)**.** *Use the repository normalization*

$$S_{\mathrm{red}}[g] = \int \sqrt{-g}\left(R_{\mu\nu}R^{\mu\nu} - \frac{1}{3}R^2\right) = \frac{1}{2}\int \sqrt{-g}\,(C^2 - E_4). \tag{8}$$

On a conformally flat background, modulo the Euler boundary term,

$$\boxed{B_{\mathrm{lin}} = C_1^\sharp C_1.} \tag{9}$$

Consequently,

$$B_{\mathrm{lin}}K = 0, \qquad K^\sharp B_{\mathrm{lin}} = 0, \qquad B_{\mathrm{lin}}^\sharp = B_{\mathrm{lin}}. \tag{10}$$

Proof. Since $C[\bar{g}] = 0$, the mixed second variation of (8) is

$$\delta_h\delta_k S_{\mathrm{red}} = \langle C_1 h, C_1 k\rangle_{\mathcal{W}}. \tag{11}$$

The definition $\delta S_{\mathrm{red}} = \langle h, \mathcal{E}(g)\rangle$ and $B_{\mathrm{lin}} = D\mathcal{E}|_{\bar{g}}$ gives $\langle h, B_{\mathrm{lin}}k\rangle = \langle C_1 h, C_1 k\rangle_{\mathcal{W}}$, proving (9). Equation (7) then gives the first detour identity, formal self-adjointness gives the second, and Hessian symmetry gives the third. □

The Weyl-fibre pairing in Lorentz signature is indefinite. Thus $C_1^\sharp C_1$ is a formal-adjoint factorization, not a positivity factorization. In trace-free conformal geometry the corresponding local sequence is

$$\Gamma(T) \xrightarrow{K_0} \Gamma(S_0^2T^*) \xrightarrow{B_{\mathrm{lin}}} \Gamma(S_0^2T^*) \xrightarrow{K_0^\sharp} \Gamma(T^*). \tag{12}$$

On Bach-flat backgrounds this is the conformal deformation detour complex of [1]. Formal self-adjointness holds in all signatures; ellipticity is a Riemannian statement and is not assumed on the Lorentzian cylinder.

2.2 The global four-dimensional deformation complex

The detour sequence (12) is not the only complex available on the conformally flat cylinder. On an oriented four-dimensional conformally flat pseudo-Riemannian manifold, the adjoint BGG deformation complex is [2, 7]

$$\Gamma(T) \xrightarrow{K_0} \Gamma(S_0^2T^*[2]) \xrightarrow{C_1} \Gamma(\mathcal{C}[2]) \xrightarrow{C_1^\sharp\star} \Gamma(S_0^2T^*[-2]) \xrightarrow{K_0^\sharp} \Gamma(T^*[-2]). \tag{13}$$

Here $\mathcal{C}[2]$ is the algebraic Weyl-curvature bundle and $C_1^\sharp\star$ means $C_1^\sharp \circ \star$, with the Hodge star acting on one antisymmetric index pair. In particular,

$$\boxed{C_1^\sharp \star C_1 = 0.} \tag{14}$$

The repository certificate verifies this identity directly as a constant-coefficient differential-operator identity in both Euclidean and Lorentzian conventions; Gover and Peterson supply the invariant global sequence.

On a locally conformally flat manifold, the flat BGG complex is a fine resolution of the sheaf of parallel sections of the flat adjoint-tractor bundle [8, 7]. The cylinder

$$M = \mathbb{R} \times S^3 \simeq S^3 \tag{15}$$

is simply connected, so that flat local system is the constant system with fibre $\mathfrak{so}(4,2)_{\mathbb{C}}$. Fine-resolution cohomology therefore gives an all-level global statement without an elliptic argument.

Theorem 2.2 (Global smooth deformation cohomology)**.** *For smooth complexified global sections on the conformal cylinder,*

$$H^q_{\mathrm{def}}(M) \cong H^q(S^3;\mathbb{C}) \otimes \mathfrak{so}(4,2)_{\mathbb{C}} \cong \begin{cases} \mathfrak{so}(4,2)_{\mathbb{C}}, & q = 0, 3, \\ 0, & q = 1, 2, 4. \end{cases} \tag{16}$$

In particular, the metric and curvature slots are globally exact:

$$\boxed{\ker C_1 = \operatorname{im} K_0,} \qquad \boxed{\ker(C_1^\sharp\star) = \operatorname{im} C_1.} \tag{17}$$

Proof. The fine resolution computes sheaf cohomology of the flat adjoint local system. Simple connectivity trivializes its monodromy, while $\mathbb{R} \times S^3$ deformation retracts onto S^3. The only nonzero ordinary cohomology groups are in degrees zero and three. Vanishing in degrees one and two is exactly the pair of statements (17). □

The theorem is signature-independent and concerns unrestricted smooth global sections. It does not by itself establish continuity, closed range, or bounded inverses on a Hilbert or Krein completion.

2.3 The Bach quotient as an on-shell curvature space

The passage from deformation-complex exactness to a detour equation is a general exact-sequence argument.

Proposition 2.3 (Detour–curvature lemma)**.** *Let*

$$E_0 \xrightarrow{K} E_1 \xrightarrow{C} E_2 \xrightarrow{D_2} E_3 \tag{18}$$

be a complex which is exact at E_1 and E_2, and let $G : E_2 \to E_1'$ be an operator such that the detour equation is $B = GC$. Then $[h] \mapsto Ch$ induces an isomorphism

$$\boxed{\frac{\ker B}{\operatorname{im} K} \cong \ker D_2 \cap \ker G.} \tag{19}$$

Proof. If $Bh = 0$, then $GCh = 0$, while $D_2Ch = 0$ because the displayed sequence is a complex. If $Ch = 0$, exactness at E_1 gives $h = K\xi$, so the induced map is injective. Conversely, if $U \in \ker D_2 \cap \ker G$, exactness at E_2 gives $U = Ch$; then $Bh = GU = 0$, proving surjectivity. □

The middle local cohomology relevant to the free field is

$$\mathcal{H}_B := \frac{\ker B_{\text{lin}}}{\operatorname{im} K_0}. \tag{20}$$

In the four-dimensional deformation complex,

$$D_2 = C_1^\sharp\star, \qquad G = C_1^\sharp, \qquad B_{\text{lin}} = GC_1. \tag{21}$$

Define the geometrically on-shell curvature space

$$\mathcal{W}_{\text{geom}} := \ker(C_1^\sharp\star) \cap \ker C_1^\sharp. \tag{22}$$

The smooth global BGG theorem and Proposition 2.3 identify the metric quotient with this space exactly.

Theorem 2.4 (Cylinder detour–curvature isomorphism)**.** *For smooth complexified fields on the conformal cylinder, the curvature map $[h] \mapsto C_1h$ induces an $SO(4,2)$-equivariant isomorphism*

$$\boxed{\frac{\ker B_{\text{lin}}}{\operatorname{im} K_0} \cong \mathcal{W}_{\text{geom}} = \ker C_1^\sharp \cap \ker(C_1^\sharp\star).} \tag{23}$$

Proof. If $B_{\text{lin}}h = 0$ and $U = C_1h$, then the factorization (9) gives $C_1^\sharp U = 0$, while (14) gives $C_1^\sharp \star U = 0$. Thus the curvature map lands in the displayed intersection.

If $C_1h = 0$, exactness at the metric slot in (17) gives $h = K_0\xi$, proving injectivity. Conversely, suppose U lies in both kernels. Exactness at the curvature slot gives $U = C_1h$ for a smooth global h, and then $B_{\text{lin}}h = C_1^\sharp U = 0$. Hence every element of the intersection has a Bach-flat metric potential, proving surjectivity. □

In Lorentz signature, complexified Weyl tensors split into Hodge eigenspaces $U = U_+ + U_-$ with $\star U_\pm = \pm iU_\pm$. If $a = C_1^\sharp U_+$ and $b = C_1^\sharp U_-$, the two target equations are

$$a + b = 0, \qquad ia - ib = 0, \tag{24}$$

so $a = b = 0$. Define the smooth on-shell chiral geometric spaces by

$$\mathcal{W}^{\text{sm}}_{\text{geom},\pm} := \{U_\pm : \star U_\pm = \pm iU_\pm,\ C_1^\sharp U_\pm = 0\}. \tag{25}$$

Then Theorem 2.4 becomes

$$\boxed{\frac{\ker B_{\text{lin}}}{\operatorname{im} K_0} \cong \mathcal{W}^{\text{sm}}_{\text{geom},+} \oplus \mathcal{W}^{\text{sm}}_{\text{geom},-}.} \tag{26}$$

All maps commute with $SO(4,2)$. Hence the quotient-to-curvature map preserves compact energy and both rotation spins on quotient classes. It remains to match this geometric system to the abstract positive-energy module constructed in Section 3.

Theorem 2.5 (Algebraic cylinder realization)**.** *Let $\mathcal{W}^{\text{alg},+}_{\text{geom},\pm}$ denote the D-finite, $SO(4)$-finite positive-energy solutions of (25). The curvature construction restricts to an isomorphism*

$$\boxed{\mathcal{W}^{\text{alg},+}_{\text{geom},+} \oplus \mathcal{W}^{\text{alg},+}_{\text{geom},-} \cong \mathcal{W}_+ \oplus \mathcal{W}_-.} \tag{27}$$

Every curvature basis vector has an explicit Bach-flat metric representative in the same D-energy and $SO(4)$ type.

Proof. Put $r = \tan(\beta/2)$, $z = 1+r^2$, and $q = d\beta + i\sin\beta\, d\gamma$. For positive chirality the normalized highest-weight metric representatives are the radiation-gauge cylinder modes

$$\begin{array}{c|c|c} \text{tower} & J & \text{nonzero metric component} \\ \hline E_n & n/2 & h_{ij} = N_E(J)T[J]_{ij}e^{-int} \\ A_n & (n-1)/2 & h_{0i} = N_A(J)V[J]_i e^{-int} \\ L_n & (n-2)/2 & h_{ij} = N_L(J)T[J]_{ij}e^{-int}. \end{array} \tag{28}$$

Here

$$V[J] = -\frac{\sqrt{2J}}{4\pi}\cos^{2J-1}\frac{\beta}{2}\, e^{-i[(J+\frac{1}{2})\alpha+(J-\frac{1}{2})\gamma]}q,$$
$$T[J] = \frac{\sqrt{2(2J-1)}}{16\pi}\cos^{2J-2}\frac{\beta}{2}\, e^{-i[(J+1)\alpha+(J-1)\gamma]}q\otimes q, \tag{29}$$

and

$$N_E = \frac{1}{4\sqrt{J(2J+1)}}, \qquad N_A = \frac{1}{2\sqrt{(2J-1)(2J+1)(2J+3)}}, \qquad N_L = \frac{1}{4\sqrt{(J+1)(2J+1)}}. \tag{30}$$

An exact coordinate calculation retains symbolic integer n, constructs all components of $C_1 h$, and applies $C_1^\sharp$ independently. After suppressing the common factor $e^{-i(nt+m_L\alpha+m_R\gamma)}z^{-a}$, nonzero pivots are

$$(C_1h_E)_{0202} = \frac{\sqrt{n(n-1)(n+1)}}{16\pi z^2},$$
$$(C_1h_A)_{0102} = \frac{\sqrt{n(n-2)(n-1)}}{32\pi\sqrt{n+2}\, z}, \tag{31}$$
$$(C_1h_L)_{0202} = \frac{\sqrt{n(n-3)(n-1)}}{16\pi z^2}.$$

They are nonzero at every allowed tower energy. The full tensors are algebraically trace-free, have one common Hodge chirality, and obey $C_1^\sharp C_1 h = 0$ identically in n. Naturality of C_1 extends each nonzero highest-weight intertwiner to its complete $SO(4)$ irrep. Defining $U_{F,n,M} = C_1 h_{F,n,M}$ therefore gives the explicit same-block inverse

$$R_n U_{F,n,M} = h_{F,n,M}, \qquad C_1 R_n = \mathrm{id}, \qquad F \in \{E, A, L\}. \tag{32}$$

The orientation-reversing cylinder isometry $\alpha \leftrightarrow \gamma$ exchanges the two $SU(2)$ factors and supplies the other chirality.

For exhaustion, the flat positive-chirality BGG data consist of the lowest-weight curvature primary $(2; 2, 0)$, the equation module $(4; 1, 1)$, and its identity $(5; \frac{1}{2}, \frac{1}{2})$. Their alternating character is

$$\chi_{(2;2,0)} - \chi_{(4;1,1)} + \chi_{(5;\frac{1}{2},\frac{1}{2})} = \frac{5q^2 - 9q^4 + 4q^5}{(1-q)^4}. \tag{33}$$

The cylinder module $\mathcal{W}_+$ is cyclic from the same primary and has this character by (110); parity supplies the other chirality. The nonzero graded intertwiner and equality of every graded $SO(4)$ multiplicity prove exhaustion. Equations (28)–(32) supply finite harmonic metric representatives directly, excluding secular or infinite-harmonic preimages in this algebraic category. The exact coordinate tensor calculation, a machine-readable certificate, and the generated pivot table are produced by `symbolic/verify_conformal_cylinder_preimages.py`. □

2.4 The finite-jet calculation as an independent convention audit

The earlier finite-jet calculation remains useful, but it is no longer a substitute for an unknown all-level theorem. On homogeneous Euclidean polynomial jets of degree $n = 2, \ldots, 6$, the matrices for K, C_1, and B_1 obey

$$C_1 K = 0, \qquad B_1 K = 0, \qquad \ker C_1 = \operatorname{im} K. \tag{34}$$

The exact ranks are shown in Table 1.

n	$\dim G_n$	$\dim h_n$	$\dim W_{n-2}$	$\dim B_{n-4}$	rank K	rank C_1	rank B_1	$\dim(\ker B_1/\operatorname{im} K)$
2	90	100	10	0	90	10	0	10
3	160	200	40	0	160	40	0	40
4	259	350	100	10	259	91	9	82
5	392	560	200	40	392	168	32	136
6	564	840	350	100	564	276	74	202

Table 1: Exact homogeneous-polynomial detour ranks. The finite Euclidean jet calculation independently audits the conventions and the first five levels of the global BGG theorem.

The quotient dimensions $(10, 40, 82, 136, 202)$ agree with the first five levels of the parity-complete $E/A/L$ module introduced below. The same calculation separates a 15-dimensional kernel of the low-degree gauge map, generated by four translations, six rotations, one dilation, and four special conformal transformations. These finite results are independent regression rails for (17) and (27) in the repository normalization.

2.5 The finite global zero-mode pair

The same topology that kills the metric and curvature cohomology leaves two finite-dimensional groups:

$$H^0_{\mathrm{def}}(M) \cong \mathfrak{so}(4,2)_{\mathbb{C}}, \qquad H^3_{\mathrm{def}}(M) \cong \mathfrak{so}(4,2)_{\mathbb{C}}. \tag{35}$$

The degree-zero copy is the familiar space of fifteen conformal Killing fields. The degree-three copy lies on the adjoint equation side of the self-dual deformation complex. Refining both groups by compact weight gives

$$4_{-1} \oplus 7_0 \oplus 4_{+1}. \tag{36}$$

Poincaré duality on S^3 and the Killing form pair the two copies, with

$$\kappa(\mathfrak{g}_r, \mathfrak{g}_s) = 0 \qquad \text{unless} \qquad r + s = 0. \tag{37}$$

Its dimension, representation, and position are exactly those expected of the dual residual constraint sector. It is therefore a strong candidate for a geometric realization of the fifteen moment-map or Taub components constructed in Section 4. The full five-term BGG sequence must nevertheless be kept distinct from the shorter four-term Bach/BV detour complex. Proposition 8.5 below derives the endpoint duality of the latter directly from cyclic finite-dimensional linear algebra, without identifying it by fiat with BGG H^3. Locating the degree-three BGG bundle representative remains a separate geometric comparison, not a premise of the BV endpoint theorem.

Thus the smooth deformation complex contains a controlled finite pair, not an unspecified collection of extra global classes. The shorter BV detour chain has its own exact endpoint quotient, while comparison of the two realizations remains useful but is not needed for the residual rank calculation.

2.6 Energy-mode Krein completion and its boundary

The smooth and algebraic theorems above now admit a natural completion in the positive-frequency cylinder-energy variables used by the residual calculation. This is deliberately narrower than a covariant completion of the metric Bach/BV complex.

Let

$$\mathcal{W}_{\mathrm{alg}} = \bigoplus_{n\geq 2}^{\mathrm{alg}} \mathcal{H}_n \tag{38}$$

be the parity-complete $E/A/L$ module. Declare its normalized cylinder basis orthonormal for a positive Hilbert majorant and complete the direct sum to $\mathscr{H}_1$. The block involution

$$\mathfrak{J}_1 = +1_E \oplus (-1_A) \oplus (-1_L) \tag{39}$$

is a bounded self-adjoint involution of norm one, and $[u, v]_1 = (u, \mathfrak{J}_1 v)_0$ extends the algebraic invariant form. Both the positive and negative eigenspaces are infinite-dimensional. Thus this is an infinite-index Krein space, not a Pontryagin space.

The reduced proper-conformal coefficients obey an all-level linear bound $|c_{\alpha\beta}(n)| \leq 2(1+n)$. With the unit-normalized Clebsch–Gordan maps, $K^{\pm}$ are finite-band matrix-weighted shifts of energy order one. Their finite-support realizations are closable, energy truncations are graph cores, and their closures equal the maximal square-summability domains. The equality

$$(\overline{K^-})^{\sharp} = \overline{K^+} \tag{40}$$

includes equality of those maximal domains. The conformal commutators are claimed on the common finite-energy invariant core only; no exponentiation to a global Krein-unitary $SO(4,2)$ representation is inferred [6].

Second quantization gives the Hilbert Fock space

$$\mathscr{F} = \Gamma_s(\mathscr{H}_1), \qquad \mathfrak{J}_{\mathscr{F}} = \Gamma_s(\mathfrak{J}_1), \tag{41}$$

and hence a bosonic Krein–Fock completion. Give the finite residual ghost exterior algebra its positive monomial Hilbert norm and set $\mathscr{K} = \mathscr{F} \widehat{\otimes} \Lambda^\bullet \mathfrak{g}^*$. The centered ghost insertion remains a bounded complementary-degree cohomological pairing; it is not promoted here to a nondegenerate Krein metric on every ghost-number block.

Theorem 2.6 (Energy-mode Krein completion)**.** *For the selected closed-cylinder problem in which all fifteen residual generators, including D, are gauged, the algebraic residual differential has a closed densely defined maximal extension $\overline{Q}$ on $\mathscr{K}$, and the algebraic state space is a graph core. Every total compact-degree eigenspace $\mathscr{K}_\delta$ is finite-dimensional. On the complement of $\mathscr{K}_0$, the bounded operator*

$$h = \bigoplus_{\delta \neq 0} \frac{\iota_D}{\delta} \tag{42}$$

preserves $\operatorname{Dom} \overline{Q}$ *and satisfies*

$$\overline{Q} h + h \overline{Q} = 1 - P_0. \tag{43}$$

The range of $\overline{Q}$ is closed, ordinary and reduced cohomology coincide, and

$$\boxed{H^4(\mathscr{K}, \overline{Q}) = \operatorname{span}\{[W_+^2], [W_-^2]\}, \qquad G_{\text{completed}} = I_2.} \tag{44}$$

Proof. The ghost compact spectrum is contained in $[-4, 4]$. At fixed total degree δ, matter energy therefore belongs to a finite interval. Fixed matter energy bounds particle number by $N \leq E/2$, uses only the finite one-particle blocks with $n \leq E$, and has finitely many bosonic partitions. Thus every $\mathscr{K}_\delta$ is finite-dimensional. In particular, $\mathscr{K}_0$ has matter energy at most four and is literally the algebraic centered block.

Write Q_δ for the finite nilpotent matrix on $\mathscr{K}_\delta$. The maximal Hilbert direct sum, with domain $\sum_\delta \|Q_\delta \Psi_\delta\|^2 < \infty$, is closed; total-degree truncations converge in norm and graph norm. Blockwise nilpotency also places $\overline{Q}\Psi$ back in the domain and gives $\overline{Q}^2 \Psi = 0$.

Exterior contraction ι_D is bounded of norm one, and nonzero δ is integral, so $\|h\| \leq 1$. The estimate

$$\|Q_\delta h_\delta \Psi_\delta\| \leq \|\Psi_\delta\| + \|h_\delta\| \, \|Q_\delta \Psi_\delta\| \tag{45}$$

proves domain invariance and extends the algebraic Cartan identity to the maximal domain. Hence every off-center closed vector is exact, and the off-center range equals the closed kernel. The centered range is finite-dimensional, so the total range is closed. The already-certified finite centered calculation then applies without a limiting argument and gives the displayed cohomology and Gram matrix. □

This theorem completes modes of the already-selected polarized Weyl state complex. It does not by itself identify a metric-field Cauchy topology. That reduced one-particle comparison is supplied next. The theorem still does not make the choice to gauge D universal: if D is retained as a Hamiltonian or boundary charge, the Cartan contraction is not part of that physical complex.

2.7 Lorentzian Cauchy–Sobolev realization

The energy-mode completion can now be reached directly from reduced metric fields on the Lorentzian cylinder. Let D_i denote the Levi–Civita connection of the unit S^3. On transverse-traceless spatial tensors and transverse one-forms define

$$(\mathcal{C}_2 h)_{ij} = \epsilon_i{}^{kl} D_k h_{lj}, \qquad (\mathcal{C}_1 v)_i = \epsilon_i{}^{jk} D_j v_k. \tag{46}$$

Exact constant-curvature index algebra gives

$$\boxed{\mathcal{C}_2^\dagger = \mathcal{C}_2, \qquad \mathcal{C}_2^2 = -D^2 + 3; \qquad \mathcal{C}_1^\dagger = \mathcal{C}_1, \qquad \mathcal{C}_1^2 = -D^2 + 2.} \tag{47}$$

The tensor curl preserves symmetry, trace and divergence. The curvature terms are fixed by $[D^n, D_i]h_{nj} = 3h_{ij}$ and $[D^n, D_i]v_n = 2v_i$ in our convention.

Using (47), the Euclidean Weyl-graviton operator of Ref. [15] continues to the local Lorentzian factorization

$$\boxed{B_{\mathrm{TT}} = P_- P_+, qquad P_\pm = \partial_t^2 + (\mathcal{C}_2 \pm 1)^2 = \partial_t^2 - D^2 + 4 \pm 2\mathcal{C}_2.} \tag{48}$$

Each factor has wave principal part and a first-order curl term. A local extension to spatial two-tensors is normally hyperbolic, its TT subspace is reducing, and the restriction supplies the reduced TT Green operators. The transverse-vector branch is

$$P_A = \partial_t^2 + \mathcal{C}_1^2 = \partial_t^2 - D^2 + 2, \tag{49}$$

and is normally hyperbolic as well. Closure of Green-hyperbolic operators under composition and direct sum then makes the reduced physical TT-plus-vector system Green hyperbolic [5]. For $B_{\mathrm{TT}} = P_- \circ P_+$, the ordered Green operator is $G^{\pm}_{P_+} G^{\pm}_{P_-}$.

The local factors interchange the two frequencies between tensor helicities. For polarization define the nonlocal spectral operators

$$A_E = |\mathcal{C}_2| - 1, qquad A_L = |\mathcal{C}_2| + 1, qquad A_A = |\mathcal{C}_1|. \tag{50}$$

On a TT harmonic labelled by $r \geq 0$, $|\mathcal{C}_2| = r + 3$, so the frequencies are $r + 2$ and $r + 4$. On the metric-derived vector branch $|\mathcal{C}_1| = r + 2$ with $r \geq 1$: the $r = 0$ band consists of Killing vectors whose symmetrized gradient vanishes. Hence the physical A tower starts at energy three, not two. At compact energy N the exact parity-complete multiplicities are

$$\boxed{d_E(N) = 2(N-1)(N+3), \quad d_A(N) = 2(N-1)(N+1), \quad d_L(N) = 2(N-3)(N+1),} \tag{51}$$

on the respective ranges $N \geq 2, 3, 4$. Thus E, A, and L are exactly the lower TT, non-Killing transverse-vector, and upper TT field branches, not merely character-matched modules.

The equation factorization alone does not determine the topology. The quadratic Weyl action does. In the convention $S = -\alpha_g \int C^2$, $\alpha_g > 0$, its TT Pais–Uhlenbeck and reduced vector symplectic forms give the positive residue magnitudes

$$R_E = R_L = 4|\mathcal{C}_2|, \qquad R_A = 2(A_A^2 - 4), \tag{52}$$

with signs $(+, -, -)$. In particular R_A has elliptic order two, not zero. The positive-frequency coordinate

$$z_\alpha = \frac{1}{\sqrt{2}} \left[(R_\alpha A_\alpha)^{1/2} q_\alpha + i (R_\alpha A_\alpha^{-1})^{1/2} p_\alpha \right] \tag{53}$$

therefore induces the Cauchy space

$$\boxed{(H^1_{\mathrm{TT}} \oplus L^2_{\mathrm{TT}})_E \oplus (H^{3/2}_{\mathrm{T}} \oplus H^{1/2}_{\mathrm{T}})_A \oplus (H^1_{\mathrm{TT}} \oplus L^2_{\mathrm{TT}})_L.} \tag{54}$$

The normalized metric waves obey $2R_\alpha(N)N|N_\alpha|^2 = 1$ at every level. Together with (51), this proves the following result.

Theorem 2.7 (Cylinder Cauchy–Sobolev realization)**.** *The positive-frequency harmonic transform from the reduced branch Cauchy space* (54) *extends by density to a Krein-unitary isomorphism onto the one-particle energy-mode completion $\mathscr{H}_1$ of Section 2.6. It intertwines the field-induced signs with $\mathfrak{J}_1 = +1_E \oplus (-1_A) \oplus (-1_L)$. The reduced TT Bach and vector operators are Green hyperbolic through* (48) *and* (49)*.*

For raw TT data (h_0, h_1, h_2, h_3), branch extraction uses $(4|\mathcal{C}_2|)^{-1}$ and defines a mixed pullback graph norm. We do not replace it by an unsupported product-Sobolev norm.

2.8 Curved auxiliary identities and the Weyl-curvature route

Editorial note for the major-revision split. The propositions and causal-transport theorem in this subsection are the mathematical core of the causal companion paper, since published as Paper VIII (`paper/08-conformal-covariant-causal-transport.pdf`). The intervening coefficient-search chronology, discarded witness families, rank screens and implementation ledgers are retained here to keep this archival monolith reproducible, but are marked for the computational supplement in the split described by `notes/conformal-paper-split-roadmap.md`.

Theorem 2.7 concerns the reduced physical branches. The all-degree curved operator statement and local BV equivalence are obtained through an ordinary-derivative realization. On the complete trace-free metric bundle define

$$T = \Box\delta - \frac{1}{3}d\delta^2 + \frac{R}{3}\delta - \mathrm{Ric}\circ\delta + \frac{1}{3}d\langle \mathrm{Ric}, h\rangle. \tag{55}$$

Exhaustive finite-order jet algebra on the homogeneous cylinder gives the global natural-operator identities

$$\boxed{TK = \Box(\Box + 2), \qquad B_{\mathrm{lin}}K = 0.} \tag{56}$$

The first operator factors as $(\Box - \mathrm{Ric} + 2)(\Box + \mathrm{Ric})$, since $\nabla\,\mathrm{Ric} = 0$ and $\mathrm{Ric}^2 = 2\,\mathrm{Ric}$. Both factors are normally hyperbolic. If $H = B_{\mathrm{lin}} + \frac{1}{2}KT$, then $HK = \frac{1}{2}K\Box(\Box + 2)$ follows formally. With the action normalization, the graded self-adjoint witness blocks are $(T, 2\sharp^{-1}, T^\sharp)$, so the actual field block of $QW + WQ$ is $2H = 2B_{\mathrm{lin}} + KT$.

A direct product of H on every component of $S^2_0T^*M$ has not been certified. The local auxiliary construction instead uses the ordinary-derivative conformal-gravity action of Ref. [16], with fields (h_{ab}, f_{ab}, v_a). Its modified de Donder conditions obey

$$\delta C_{0a} = \Box\xi_{-2,a} - \xi_{0,a}, \qquad \delta C_{2a} = \Box\xi_{0,a}, \qquad \delta C_1 = \Box\sigma. \tag{57}$$

Thus the combined ghost principal symbol is $g^{\mu\nu}\zeta_\mu\zeta_\nu I_9$. Direct reconstruction from the exact covariant action and gauge fermion gives the complete curved Hessian E_{aux}, gauge map K_{aux}, companion, and four-row operators. In a canonical covariant derivative normal form they obey

$$E^\sharp_{\mathrm{aux}} = E_{\mathrm{aux}}, \qquad E_{\mathrm{aux}}K_{\mathrm{aux}} = 0, \qquad Q^2_{\mathrm{aux}} = 0, \tag{58}$$

and

$$Q_{\text{aux}}W_{\text{aux}}+W_{\text{aux}}Q_{\text{aux}}=P_{\text{aux}},\qquad W_{\text{aux}}^{\sharp}=W_{\text{aux}},\qquad P_{\text{aux}}^{\sharp}=P_{\text{aux}}. \tag{59}$$

The coefficient reconstruction passes 630/630 exhaustive order-three and order-four cylinder jet vectors and globalizes by homogeneity. These exact operator identities are what the repository flag `curved_operator_identity` records. They must not be confused with normal hyperbolicity of the field block.

Indeed, the latter scalar-wave target is impossible for this field and gauge content. At the null covector $\zeta=(1,1,0,0)$ exact nonzero-minor certificates give

$$\boxed{\operatorname{rank}E_2(\zeta)=11,\qquad \operatorname{rank}K_1(\zeta)=9.} \tag{60}$$

For every invertible pointwise fibre map J, one has $\operatorname{rank}(J^{-1}E_2)=11$. For every first-order companion C, one has $\operatorname{rank}(K_1C_1)\le 9$. At a null covector a scalar normally hyperbolic symbol would vanish, so the two terms cannot cancel. This proves a no-go theorem for the proposed 24-component pointwise scalar-wave witness, not for Green hyperbolicity.

The same symbol calculation explains the rank defect. With $N_2=J_{\text{act}}^{-1}E_2$,

$$\operatorname{im}N_2/\operatorname{im}K_1\cong\mathcal{H}_{+2}\oplus\mathcal{H}_{-2},\qquad \dim_{\mathbb{C}}(\operatorname{im}N_2/\operatorname{im}K_1)=2. \tag{61}$$

In a real basis this quotient is represented by $f_{22}-f_{33}$ and f_{23}; the transverse $SO(2)$ generator squares to $-4I_2$. The linearized Weyl symbol descends to the Hessian-reduced double quotient

$$\frac{F/\ker N_2}{N_2^{-1}(\operatorname{im}K_1)/\ker N_2}\longrightarrow\frac{\operatorname{im}\mathcal{W}_2}{\mathcal{W}_2(N_2^{-1}(\operatorname{im}K_1))} \tag{62}$$

and, in the certified quotient bases, is exactly

$$\boxed{\sigma(\mathcal{W})_{\text{red}}=\frac{1}{4}I_2.} \tag{63}$$

Both sides of (62) have dimension two. We do not claim $\ker\mathcal{W}_2=\operatorname{im}K_1$ on the unreduced 24-field fibre.

The local BV-canonical change

$$\eta=\xi_0-d\sigma,\qquad \widehat{f}=f-A_g^{-1}G^b(g,b) \tag{64}$$

isolates $Qv=-\eta$. The auxiliary mass map and inverse are pointwise,

$$A_g(\phi)=-\frac{1}{2}(\phi-g\operatorname{tr}_g\phi),\qquad A_g^{-1}(s)=-2s+\frac{2}{3}g\operatorname{tr}_g s, \tag{65}$$

and the transformation of every antifield row is generated by the same local BV functional. The changes, their inverses, and this elimination preserve support.

Proposition 2.8 (Curved auxiliary identities and support-local SDR)**.** *The ordinary-derivative tensor–tensor–vector BV complex on the Lorentzian conformal cylinder has exact curved four-row operator identities. The local BV-canonical change above conjugates the actual curved differential to the metric complex plus three generalized-auxiliary arrows. The resulting maps satisfy*

$$p\circ i=\mathrm{id},\qquad i\circ p-\mathrm{id}=Qk+kQ \tag{66}$$

on every field, ghost, antifield, trace/Weyl, and nonminimal row. Their entries are finite differential maps or pointwise inverses, so they preserve compact, spacelike-compact, and unrestricted smooth support. The same canonical split gives the action-level presymplectic comparison

$$i^*\omega_{\mathrm{aux}} - \omega_{\mathrm{met}} = d\beta + Q\gamma \tag{67}$$

and equality of the induced closed-S^3 Cauchy pairings on cohomology. Thus the repository flags `curved_deformation_retract` *and* `curved_current_comparison` *are true. No local conformal-Killing projector is used.*

The exact physical symbol quotient selects curvature prolongation as the current causal route. For an algebraic Weyl tensor Ψ define its electric and magnetic STF parts by

$$E_{ij} = \Psi_{0i0j}, \qquad B_{ij} = \frac{1}{2}\epsilon_i{}^{kl}\Psi_{0jkl}. \tag{68}$$

The ten-dimensional algebraic bundle $\mathrm{STF}_2(S^3) \oplus \mathrm{STF}_2(S^3)$ does not close by itself at first order. If $V_{\mu,\nu,\sigma} = \nabla^\rho\Psi_{\mu\rho\nu\sigma}$, its independent components form a sixteen-dimensional Cotton slot, naturally two trace-free 3×3 matrices. Exact contraction of the Cotton definition, Bach row, and Hodge-dual compatibility row gives a 34×26 first-order coefficient table on (E, B, V) whose temporal matrix has rank 26 and leaves eight constraints. An exhaustive comparison on all $10\binom{6}{4} = 150$ Weyl two-jets proves equality with the curved covariant Bianchi–Bach equations and globalizes by cylinder homogeneity. Thus `curved_EB_equations` and `curved_EB_first_order_closure` are exact.

This result is not yet a hyperbolicity theorem. The canonical unadjusted choice of 26 evolution rows has characteristic polynomial

$$\frac{1}{144}\lambda^2(\lambda^2-1)^8(4\lambda^2-1)^2(3\lambda^2+1)^2, \tag{69}$$

so the pair $\lambda = \pm i/\sqrt{3}$ proves that constraint additions are necessary. Exact constant-curvature symbol algebra nevertheless gives

$$\operatorname{div}\operatorname{curl}_2 = \frac{1}{2}\operatorname{curl}_1\operatorname{div}. \tag{70}$$

The constraint-adjusted square system is now exact at the level required for the Cauchy problem. Its first twenty rows are covariant combinations, while its final six differ from the vector Bach rows by the two vector constraints a, c. Consequently the pointwise row modules differ by rank six. This is not a solution-space defect: differentiating the primary q, r constraints and using the exact vector Bach rows generates $a = c = 0$. Conversely, the adjusted rows together with all fourteen constraints recover every covariant row. Thus the two presentations generate the same formally integrable differential ideal.

The adjusted 26-state system has a positive symmetrizer and causal speeds $0, \pm\frac{1}{2}, \pm 1/\sqrt{3}, \pm 1$. Its complete sourced subsidiary identity includes the unit-S^3 curvature commutator correction, and compatible sources preserve all fourteen constraints. This proves the three flags

`curved_EB_symmetric_hyperbolicity`,
`curved_sourced_constraint_identity`, and
`curved_constraint_propagation`.

The unadjusted raw subsidiary system retains an imaginary longitudinal pair; the certified local constraint addition, not a hidden spectral projection, removes it.

The global solution-space audit is independent of that analytic repair and is already exact. The exhaustive jet equivalence, the smooth cylinder BGG theorem, and the symbolic all-energy metric preimages give

$$\ker(C_1^\sharp) \cap \ker(C_1^\sharp \star) \cong E \oplus A \oplus L \tag{71}$$

with both chiralities. Symbolically, $\dim \mathcal{C}_n - 2\,\mathrm{rank}\, C_1^\sharp = 6n^2 - 14$, equal to the parity-complete $E/A/L$ multiplicity, and the exact rational characters agree. The Cotton definition has an algebraic identity coefficient, so its graph adds no second solution copy. Thus the flag

`EAL_curvature_spectrum_match`

is true; the levels 2 through 6 are retained only as regressions, not as the proof.

The target is not a second residual Chevalley–Eilenberg calculation. That calculation and its pairing are already complete. It is a local chain of pairing-compatible, support-preserving quasi-isomorphisms

$$\Gamma_c(\mathcal{C}_{\mathrm{met}})[1] \simeq \Gamma_c(\mathcal{C}_{\mathrm{aux}})[1] \simeq \Gamma_c(\mathcal{C}_{\mathrm{prol}})[1] \simeq \Gamma_{sc}(\mathcal{C}_{\mathrm{prol}}) \simeq \mathcal{C}_\Sigma \simeq \mathcal{W}_+ \oplus \mathcal{W}_- \quad \text{with residual endpoints.} \tag{72}$$

The curvature variable must be adjoined as a contractible prolongation, for example through $\widehat{\Psi} = \Psi - C_1 h$, rather than replacing the metric potential by curvature. Its inclusion, projection, and homotopy must contain every field, equation, identity, antifield, trace/Weyl, nonminimal, and first-order-reduction row and use only finite differential operators and pointwise inverses.

Two further support-local subtheorems are now exact. First, adjoining the graphs $\Psi = C_1 h$ and $c = \mathrm{div}\,\Psi$ with their cotangent lifts gives a BV-canonical contractible SDR. Second, the curvature rows form the autonomous compatibility complex

$$U_{26} \xrightarrow{(L,K)} F_{26} \oplus C_{14} \xrightarrow{(-R,S)} I_{14}, \qquad SK = RL, \tag{73}$$

together with its formal cotangent-adjoint complex. A zeroth-order backward witness makes $QW + WQ$ diagonal with the symmetric-hyperbolic blocks L and S and algebraic graph identities.

These subtheorems cannot be direct-summed: the autonomous curvature complex carries the nonzero $E/A/L$ module and would duplicate physical cohomology. The state/equation part of the required mapping-cylinder chain map has been constructed. It supplies the explicit local operator A_{eq} satisfying

$$(L, K)T = A_{\mathrm{eq}} E_{\mathrm{aux}}, \qquad T = (C_1, \mathrm{div}\, C_1). \tag{74}$$

The first square is verified on all $10\binom{8}{4} = 700$ metric four-jets. The formerly reported order-zero identity-square defect was a coordinate artifact. The common-Rees diagnostic had composed the curved attachment with the flat-Fourier equation projection, whereas the fibre-identified curved cotangent row uses

$$p_E = (D^{-1} S_h^\sharp D, 1, 0) \tag{75}$$

and the corresponding identity projection p_I. Reconstructing these maps coefficientwise gives

$$A_{\mathrm{eq}} = A_{\mathrm{core}} p_E, \qquad B_{\mathrm{id}} = B_{\mathrm{core}} p_I, \qquad N_{\mathrm{curv}} A_{\mathrm{eq}} = B_{\mathrm{id}} C_{\mathrm{aux}}. \tag{76}$$

The last equality follows from the exact metric-core square together with the equation-to-identity block $p_I C_{\mathrm{aux}} = C_{\mathrm{core}} p_E$ of the certified all-row curved retract. Thus the old rank-four matrix is retained only as a scoped mixed-coordinate diagnostic and is not an operator obstruction. The preceding square $TK_{\mathrm{aux}} = 0$ remains independently exhaustive on all diffeomorphism and Weyl-scalar jets and all four boost ghosts.

Substitution into the complete sixteen-block cotangent mapping cylinder now proves coefficientwise

$$Q_{\mathrm{prol}}^2 = 0, \qquad PI = 1, \qquad IP - 1 = Q_{\mathrm{prol}}H + HQ_{\mathrm{prol}}, \tag{77}$$

as well as odd BV cyclicity. Every field, equation, identity, cotangent and contractible cone row is enumerated, and I, P, H contain only finite-order differential operators and pointwise inverses. Hence `support_local_prolongation_retract` and `prolonged_BV_operator_identity` are true. The prolonged current certificate now binds the corrected curved core-chain provenance and proves $I^*\omega_{\mathrm{prol}} - \omega_{\mathrm{aux}} = d\beta + Q\gamma$ off shell, with $\beta = \delta Y_I$ and $\gamma = 0$. Hence `prolonged_current_comparison` is true. Cyclicity of the direct causal homotopies makes $\Delta_\Lambda = \Lambda_{\mathrm{prol},+} - \Lambda_{\mathrm{prol},-}$ graded antisymmetric; the common algebraic and trace contractions cancel from this difference. Combining the resulting parent Green/current identity with the prolonged-to-auxiliary and auxiliary-to-metric $d+Q$ improvements proves that the causal and Cauchy-current pairings agree on cohomology, with all-energy normalization $+I_E \oplus (-I_A) \oplus (-I_L)$.

The two legacy analytic exceptions are $E_{\mathrm{aux}} + KC$ on the auxiliary field row and its cotangent copy; they retain the scalar-wave null-rank obstruction. Thus mere direct-sum conjugation does not produce the superseded canonical prolonged witness. They do not obstruct the direct tractor causal homotopy or its pairing transport.

Two exact fail-closed diagnostics further delimit this open construction. For $q = g^{-1}(\zeta, \zeta)$, the generic prenormal principal symbol $P_2 = J_{\mathrm{act}}^{-1}E_2 + K_1C_1$ obeys

$$(P_2 - qI_{24})^2 = 0, \qquad (2qI_{24} - P_2)P_2 = P_2(2qI_{24} - P_2) = q^2 I_{24}. \tag{78}$$

Its Smith multiplicities are 6/12/6, corresponding respectively to algebraic, wave and biwave factors. This is a principal-symbol identity, not a Green theorem. Indeed, the naive frozen lower-order completion $2qI - P$ has nonzero remainders in orders zero, one and two. This rejects only that literal completion; it does not rule out corrected first-order factors or a full covariant composition.

The complete invariant lower-order ansatz makes the next gate finite. The $SO(3)$ cylinder-holonomy decomposition gives

$$\dim D_0 = 38, \qquad \dim D_1 = 93, \qquad D = D_{\mathrm{naive}} + X_1^\mu \nabla_\mu + X_0. \tag{79}$$

The exact cubic divisibility equations for both DP and PD have a simultaneous 45-parameter solution family. There is therefore no cubic obstruction. Restoring the two independent factor splittings and all algebraic coefficients gives

$$45 + 2 \cdot 93 + 5 \cdot 38 = 421 \tag{80}$$

nonlinear unknowns after the cubic gate. The scalar principal normalization qI_{24} loses no invariant solutions, since a parallel invertible qH/qH^{-1} pair may be redistributed between the two factors. The complete 421-variable problem is now assembled as an exact sparse quadratic symmetrized-PBW system. Its order-zero, order-one and order-two row counts are respectively 240, 960 and 2484. At order two, the 2484×190 matrix multiplying the algebraic variables has rank 100, so its cokernel has dimension 2384 and 90 algebraic variables remain free. Exact

Schur projection gives 2130 nonzero polynomial constraints, 365 up to scale, and no constant polynomial obstruction. These projected equations and the lower orders have not been solved.

A convention audit corrected the covector commutator: its raised curvature index uses the spatial projector, not a four-dimensional Kronecker delta, so all mixed time–space curvature components vanish. Independent coordinate-jet commutator, Weitzenböck and focused $\Box^2$ composition tests now pass. Exact triangular symmetrized-jet PBW inversion exhausts all 1680 four-jet basis elements, passes 504 ordered-word round trips and certifies associativity, so the quadratic-factor composition backend is ready. A naive transpose/reversal of the already sorted $\Box^2$ component table nevertheless leaves a 48-entry defect. This is not an independent primal-composition counterexample: the sorted coefficient matrices no longer individually carry their derivative-index slots. It does show that the naive table adjoint is invalid. The required pairing-aware adjoint has now been constructed on the symmetrized-PBW coefficient tensors. With the action pairing it verifies

$$P^\sharp = P, \qquad D^\sharp_{\text{naive}} = D_{\text{naive}}, \qquad (DP)^\sharp = PD. \tag{81}$$

Requiring $D^\sharp = D$ reduces the invariant first-order family from 93 to 44 dimensions and its algebraic family from 38 to 24. The resulting cubic matrix has size 11520×137, rank 116, and a 21-dimensional kernel. Taking $R_- = L_+^\sharp$ and $R_+ = L_-^\sharp$ makes the right factorization the adjoint of the left rather than an independent system. Restoring the left splitting and algebraic coefficients leaves a 214-parameter nonlinear problem. Its complete order-two gate has 1242 rows; the 1242×100 algebraic matrix has rank 52, leaving 48 free algebraic directions. Exact cokernel projection gives 1050 nonzero constraints, 179 up to scale, with no constant polynomial obstruction. The projected quadratic constraints and orders one and zero remain unsolved and unpromoted. All 179 normalized constraints have maximal degree two. Their 1497 quadratic monomials have coefficient rank 124; the resulting 55 left-kernel combinations have zero affine part. Hence exact vector-space reduction eliminates no variable, and the next step is a genuinely nonlinear ideal calculation rather than further linear algebra.

The complete bare-wave subfamily can nevertheless be decided exactly. If one factor in each of DP and PD is the literal rough Box, the four inner/outer orientations give rational systems with 159 unknowns and

$$\operatorname{rank} A = 159, \qquad \operatorname{rank}[A \mid b] = 160. \tag{82}$$

All four have the same one-row left-null certificate: the coefficient of $\nabla_{(0}\nabla_{1)}$ mapping f_{01} to f_{00} is -8 in the required right-hand side and vanishes in every correction column. This is a no-go only for the bare-$\Box$ subfamily. When both factors have nonzero first-order coefficients, the quadratic product A_-A_+ changes the order-two equations. In fact, a support-minimal rational zero-sum split using two invariant directions repairs this row together with its full $SO(3)$ orbit $f_{0i} \to f_{00}$. The old bare-wave obstruction therefore does not extend to the general family. Fixing that minimal split nevertheless produces a new scoped order-two obstruction after all 190 invariant algebraic variables are admitted:

$$\operatorname{rank} A = 100, \qquad \operatorname{rank}[A \mid b] = 101. \tag{83}$$

Its one-row left-null witness is the coefficient of $\nabla_{(0}\nabla_{1)} : h_{22} \to f_{01}$, whose required value is 16 and whose 190 variable coefficients vanish. This rules out only that fixed minimal split, not other first-order splits, the complete 421-variable system, or the 214-parameter sharp subfamily. Thus the cubic gate proves neither a factorization nor a Green theorem.

As a further exact screen, multiplying the 179 reduced sharp quadrics by constants and all 114 degree-one variables gives a sparse Macaulay matrix with 136585 monomial rows and 20585 multiplier columns. A deterministic minor over a prime avoiding every input denominator yields the rigorous characteristic-zero degree-three bound

$$12861 \leq \operatorname{rank}_{\mathbb{Q}} M_3 \leq 14136. \tag{84}$$

The upper bound is $124 \cdot 114$, from the exact dimension of the quadratic coefficient space. The full rational ranks have not been computed, so this screen decides neither membership of a nonzero constant in the ideal nor the low-degree quotient dimensions. It promotes no no-go or Green flag.

The exhaustive relative-incidence ledger contains nine possible odd-adjoint pairs. No single pair yields reciprocal coupling, and the smallest physical two-way candidate is the pair $4+5$. On the ordered core $(M_{\text{aux}}, X_U, Y_U^\sharp)$ it gives

$$\begin{pmatrix} E_{\text{aux}} + KC & R & S \\ S^\sharp E_{\text{aux}} & L_{26} & 0 \\ R^\sharp E_{\text{aux}} & 0 & L_{26}^\sharp \end{pmatrix}. \tag{85}$$

The finite-block Schur formula is exact, but its Schur complement contains the Green operators of L_{26} and $L_{26}^\sharp$ and is therefore nonlocal. Conversely, the unreduced saddle is order two, so the certified first-order symmetric-hyperbolic theorem does not apply to it. Moreover, its natural local instantiation

$$A_F = p_F A_{\text{eq}}, \qquad S = A_F^\sharp, \qquad R = A_F^\sharp J_U \tag{86}$$

has an exact balanced Douglis temporal rank obstruction:

$$\operatorname{rank} \sigma_{\text{DN}}(dt) \leq 107 < 116, \qquad \operatorname{defect} \sigma_{\text{DN}}(dt) \geq 9. \tag{87}$$

The temporal leading coefficient of the characteristic determinant therefore vanishes and no positive temporal symmetrizer exists for this realization. This is a no-go only for the smallest pair-$4+5$ ansatz with (86); it does not exclude larger relative witnesses or an additional support-local first-order prolongation.

The expanded-relative incidence audit exhausts all nine odd-adjoint pairs and the 162-dimensional Hom family determined by the declared cylinder $SO(3)$ multiplicities. It finds exactly three minimal reciprocal two-pair candidates, labelled $1+6$, $1+7$ and $2+7$. Even using all reciprocal curvature partners, the maximum cross-rank on each central 24-component auxiliary block is 21. The three missing directions are rotation scalars. Hence any ansatz which makes the entire central auxiliary diagonal subprincipal has temporal rank defect at least three. A viable expanded witness must retain or locally prolong these scalar directions. This is a scoped design theorem; coefficient tables, characteristics and a symmetrizer remain open, and no flag is promoted. An independent coefficientwise calculation constructs the three rotation generators in the actual component bases of all sixteen blocks. The nine relative Hom equations have nullities $4, 18, 4, 36, 14, 14, 22, 36, 14$, whose sum is 162. The explicit pair-$1+6$ maps, the temporal K and $N_{\text{curv}}^\sharp$ coefficients, the induced vector projector, and the retained scalar diagonal all have zero rotation-generator intertwining defect.

The three uncovered scalar coordinates are exactly h_{00}, f_{00} and v_0. The restriction of the existing KC coefficient to these directions is triangular of rank three and determinant -1. For the pair-$1+6$ candidate, explicit finite-order coefficient maps give the exact numerator identity

$$KR_1 N^{\sharp}_{\mathrm{curv}} R^{\sharp}_6 = -\Pi_{\mathrm{vector}}. \tag{88}$$

To pass from this numerator to the saddle Schur term $BD^{-1}C$, the actual 92-component curvature temporal diagonal is needed. In the normalized curvature equations and their formal-adjoint rows,

$$D(dt) = \mathrm{diag}(I_{26}, -I_{40}, -I_{26}), \qquad D(dt)^{-1} = D(dt). \tag{89}$$

Here the first-order compact-support convention $N^{\sharp}_{\mathrm{curv},\mu} = -N^{T}_{\mathrm{curv},\mu}$ is used both in the middle diagonal and in C. Consequently the exact pair-$1+6$ Schur term is $BD^{-1}C = +\Pi_{\mathrm{vector}}$, and the field Schur block is $E_{\mathrm{aux},2} + D_{\mathrm{scalar}} - \Pi_{\mathrm{vector}}$. Retaining the rank-three scalar diagonal gives an assembled 116×116 temporal Douglis symbol of exact rank 116 and determinant one. This closes the temporal-invertibility gate only. The scalar diagonal has not been lifted to a cyclic all-row witness, and the three spatial first-order coefficients of $R^{\sharp}_6$ have not yet been constructed. Hence the arbitrary-covector relative symbol itself, its characteristic polynomial, positive symmetrizer, lower-order completion and every-degree Green identities remain open. Thus no Green-theoretic flag follows from (88)–(89).

The subsequent complete $SO(3)$-equivariant audit makes this boundary sharper. The actual $R^{\sharp}_6$ coefficient equations have a 22-dimensional temporal commutant and a 46-dimensional spatial-vector commutant. After fixing the temporal normalization above, all 46 spatial parameters were retained. Nevertheless, for the intrinsic aligned pencil $Q(z) = P_{\mathrm{weighted}}((-z, 1, 0, 0))$ the vectors $a_0 = 2f_{23}$ and $a_1 = h_{23}$ obey

$$Q(1)a_0 = 0, \qquad Q(1)a_1 + Q'(1)a_0 = 0,$$

and the two parameter-sensitivity maps have exact rank zero. Hence every regular member of this fixed-temporal pair-$1+6$, cyclic -2Π family retains a length-two elementary divisor. This rules out a semisimple faithful strong linearization, and therefore a positive symmetric-hyperbolic symmetrizer, within this family. It is not a no-go for another relative incidence, a changed temporal normalization, a larger witness, or generalized Green hyperbolicity. This screen by itself promotes no causal or final covariant H^4 flag; the later curved tractor transfer supplies the causal homotopy by a different construction.

This non-semisimplicity is not itself a Green obstruction. An independent BV-row audit identifies $2f_{23}$ with the already contractible shifted auxiliary summand, while h_{23} has nonzero Weyl helicity two; the extension splits after the certified support-local auxiliary shift and dies as an extension on Q-cohomology. Moreover, on the aligned physical principal quotient the exact block is

$$\begin{pmatrix} qI_2 & 0 \\ 4\rho^2 I_2 & qI_2 \end{pmatrix}, \qquad q = \rho^2 - \tau^2,$$

whose abstract operator form $\left(\begin{smallmatrix} D & 0 \\ R & D \end{smallmatrix}\right)$ has the two-sided same-sided Green inverse $\left(\begin{smallmatrix} G & 0 \\ -GRG & G \end{smallmatrix}\right)$. Thus the remaining task is a projector-free local exact-extension construction, not removal of the physical Jordan chain. Refining the bundle support graph by the evolution/subsidiary equation-dual split leaves a reciprocal rank-34 $(h, f, C^{\sharp})$ component; the other curvature tails are triangular singletons. This aligned principal calculation does not yet promote the full physical biwave or prolonged Green flags, since the local curvature quotient and lower-order intertwining remain open. On this aligned channel the certified BV shift is explicit as well: writing $L = 1 - z^2$, it conjugates the complex block to $\mathrm{diag}(L^2, -1)$, while the witness remains the closed block $\left(\begin{smallmatrix} L & 0 \\ 4 & L \end{smallmatrix}\right)$ with exact same-sided inverse $\left(\begin{smallmatrix} G & 0 \\ -4G^2 & G \end{smallmatrix}\right)$. No inverse curl, Laplacian, TT projector or helicity projector occurs in this formula; what remains open is its support-local realization in the complete 116-row exact extension. On the exact TT-plus-shifted-auxiliary operator subcomplex one can therefore promote the scoped statements that the physical biwave block is Green hyperbolic and its Jordan extension is causal: the restricted witness is $\mathrm{diag}(B_{\mathrm{TT}}, 1, B_{\mathrm{TT}}, 1)$ and the corresponding restricted $Q\Lambda_\pm + \Lambda_\pm Q = 1$ identity is exact. These are not the complete prolonged Green statements. The rank-4 vector sector is contracted below, but the rank-14 equation-cone extension and a support-local map from arbitrary sources into its physical biwave carrier remain open. The first alternative-incidence screen is also exact: pairs 1+7 and 2+7 have a nonzero joint intrinsic-sensitivity image of rank 16. Their smallest temporally regular slices have determinant 8 and entirely real causal roots, but are non-semisimple at speeds 0 and ± 1; the first directly sensitive spatial correction does not repair either slice. This is not a no-go for the complete alternative families. The canonical 16-parameter subfamily spanning the full intrinsic sensitivity image is nevertheless uniformly non-semisimple: the zero-root valuation and kernel-dimension bounds are respectively 40 versus at most 33 for pair $1 + 7$, and 48 versus at most 47 for pair $2+7$. This still does not rule out the raw 122-parameter families or generalized Green extensions. A projector-free differential-module analysis reduces the reciprocal rank-34 component further. The six vector-gauge columns together with the $q^\sharp/r^\sharp$ subsidiary coordinates present a rank-12 invariant submodule with an exact recursive Green inverse. The rank-22 quotient contains a closed rank-8 symmetric-hyperbolic constraint quotient and an unresolved rank-14 field cokernel. Since $(C_1, \mathrm{div}\, C_1)K = 0$, curvature descends locally to this cokernel; constructing its induced biwave intertwiner and inverse is the remaining central block problem. The raw off-diagonal ideal is not nilpotent, so the inverse does not follow from a naive finite Neumann series. The remaining rank-4 vector field singleton, together with its mandatory primal/cotangent closure of rank 16, is already the shifted $(\eta, v, v^\sharp, \eta^\sharp)$ contractible summand. Replacing its witness locally gives $P_{\mathrm{vec}} = I_{16}$, $G_+ = G_- = I_{16}$ and $\Lambda_+ = \Lambda_- = W$, with both homotopy defects zero and vanishing causal propagator. Thus the unresolved analytic content of this filtration is the rank-14 curvature field cokernel and its all-row insertion. For reproducibility, the rank-14 handoff is persisted as a compact content-addressed manifest: it records the ordered quotient and filtration maps, full 26-state evolution and lower-order tables, all fourteen source constraints and subsidiary matrices, symmetrizers, BV row layout, formal adjoints and current conventions by authoritative path and hash rather than duplicating the large matrices. The rank-14 quotient has a projector-free local presentation. Raw $T = (C_1, \mathrm{div}\, C_1)$ has rank 5 and kernel rank 9, while the compatible-source kernel has rank 12, but $\mathrm{im}\, T$ is not contained in it: the defect ranks are 3 generically and 1 at the aligned null covector. Thus K_{12}/I_5 is undefined. This is distinct from the exact null Weyl split into rank-three gauge preimages and two helicities, on which the reduced Weyl map is $\frac{1}{4} I_2$.

The exact replacement is the equation cone

$$\widehat{T} = (T, E_{\mathrm{aux}}), \qquad \widehat{K} = (K_{\mathrm{state}}, -A_C), \qquad \widehat{K}\widehat{T} = K_{\mathrm{state}} T - A_C E_{\mathrm{aux}} = 0, \tag{90}$$

the constraint component of the certified chain square for $E_{\mathrm{curv}} = (L_{\mathrm{WC}}, K_{\mathrm{state}})$. Hence the bridge is an equation-level deformation retract using the auxiliary f equation and all forty curvature rows, not a direct state retraction.

The full cone, including its incoming gauge row, has degree ranks $9 \to 24 \to 50 \to 49 \to 14$. The actual curved projections p_E, p_I make its operator squares exact; the old rank-four defect used mixed coordinates. Moreover the fifteen-entry curved attachment supplies the degree-(-2) differential on the residual null PBW page $\mathbb{C}^2 \xrightarrow{d_{12}} \mathbb{C}^4 \xrightarrow{d_{23}} \mathbb{C}^2$. Both maps have rank two and an explicit homotopy gives $h_{12} d_{12} = 1$, $d_{12} h_{12} + h_{23} d_{23} = 1$, and $d_{23} h_{23} = 1$. Thus the null filtered cone is algebraically exact.

This finite-page result is not a total local inverse: $DH+HD=1$ would require $H_1K=I_9$ with $\operatorname{rank}K=5$ and, at the correctly typed upper endpoint, $[N,B]H_4=I_{14}$ with $\operatorname{rank}[N,B]=13$. The certified composite SDR instead gives $P_{\mathrm{alg}}=DH_{\mathrm{alg}}+H_{\mathrm{alg}}D$ as a support-local cyclic idempotent chain projector, with $P_{\mathrm{end}}=1-P_{\mathrm{alg}}$. It contracts 356 prolonged components and retains the 30-component metric–curvature graph. The endpoint has now also been reconstructed coefficientwise in the ordered rows

$$G_{\mathrm{met}}[5]\xrightarrow{K_{\mathrm{met}}}h[10]\xrightarrow{\bar{B}}\bar{E}_{\mathrm{met}}[10]\xrightarrow{C_{\mathrm{met}}}I_{\mathrm{met}}[5],\qquad \operatorname{ord}(K_{\mathrm{met}},\bar{B},C_{\mathrm{met}})=(1,4,1).\tag{91}$$

All curved lower-order coefficients are emitted. Direct coefficient checks give $Q_{\mathrm{end}}^2=0$, the two formal-adjoint identities and odd cyclicity; the local graph maps satisfy $p_{\mathrm{end}}j_{\mathrm{end}}=1$ and strictly intertwine Q_{prol} and Q_{end}. Thus the earlier $66\to30$ flat-Fourier hashes remain only a dimension/order/sign regression and are no longer the source of the curved endpoint claim.

The endpoint equation coordinates already contain the fibre identification $\bar{B}=J_{\mathrm{met}}^{-1}E_{\mathrm{Bach}}$. Consequently the abstract minimal witness coefficient $2\sharp^{-1}$ must not be inserted a second time. The coefficientwise canonical backward map is

$$W_0=(T_{\mathrm{gf}},1,T_{\mathrm{gf}}^{\sharp}),\qquad D_{\mathrm{end}}=Q_{\mathrm{end}}W_0+W_0Q_{\mathrm{end}}.\tag{92}$$

With the superseded middle coefficient $2I$, the trace-free principal matrix at $\zeta=dt$ has spectrum $1^4\oplus2^5$; this is the exact normalization counterexample missed by the first textual guard. With the identity in (92), exhaustive fourth-order coefficients instead give

$$\sigma_4(D_{\mathrm{TF}})(\zeta)=\big(g^{\mu\nu}\zeta_\mu\zeta_\nu\big)^2I_9.\tag{93}$$

The ghost block is a triangular extension of $\Box(\Box+2)I_4$ by the Weyl-scalar identity; the metric trace is likewise a triangular identity extension. Thus the sole primal analytic block is the rank-nine $D_{\mathrm{TF}}=P_{\mathrm{TF}}D_MP_{\mathrm{TF}}$, together with its formal-adjoint copy in the equation row.

The canonical curvature backward-map audit identifies a precise boundary. The upper row lifts locally: there is a rank-four algebraic map R with

$$A_{\mathrm{core}}R=i_CB_{\mathrm{core}}.\tag{94}$$

The analogous canonical middle lift would require $T_{\mathrm{core}}S=A_F$. But the exact factorization $T_{\mathrm{core}}=J_{\mathrm{WC}}W_{EB}$ obeys $\pi_{EB}J_{\mathrm{WC}}=1$, whereas $\pi_{EB}A_F=0$ and $\operatorname{rank}A_F=5$. Hence such an S is impossible on the leading Douglis page. Equivalently, the two rank-five symbol images have zero intersection on the certified generic, timelike, spacelike, null and temporal representatives. This is a no-go only for restricting the canonical zeroth-order middle map to the endpoint graph, not for a relative witness.

Indeed, the rank-five A_F incidence has an exact support-local cyclic lift through the algebraic complement of the full 386-component complex. Its two off-diagonal legs are nonzero and fixed by the $386\to66\to30$ projectors. However the minimal projected saddle has the stronger exact identity

$$P_{\mathrm{end}}L_{A_F}P_{\mathrm{alg}}L_{A_F}P_{\mathrm{end}}=0.\tag{95}$$

Thus its Schur endpoint is unchanged: $S_{\mathrm{end}}=D-CB=D$. The relative incidence introduces no new endpoint obstruction, but it also cannot improve Green invertibility of the exact thirty-row diagonal. Larger relative witnesses are not excluded.

The most direct same-bundle factorization of D_{TF} is now excluded exactly. Write

$$L_\pm = P_{\mathrm{tr}} + \Box P_{\mathrm{TF}} + A_\pm^\mu \nabla_\mu + B_\pm. \tag{96}$$

On $S_0^2 T^*M \cong V_0 \oplus V_1 \oplus V_2$, the complete parallel $SO(3)$-invariant trace-free-preserving families have dimensions nine at first order and three at order zero. A general nondegenerate wave-type leading pair qH_-, qH_+ loses no solutions under the displayed scalar normalization: the scalar target forces $H_-H_+ = 1$, and replacing the factors by L_-H_+ and $H_+^{-1}L_+$ preserves their product and the complete lower-order families. Exact symmetrized PBW composition of $L_-L_+ - D_{\mathrm{TF}}$ has no order-four defect. Order three has coefficient and augmented rank nine and forces $A_+ = -A_-$. At order two the three sums $B_- + B_+$ are fixed; the remaining 42 quadratics admit twelve explicit degree-two rational multipliers w_i satisfying

$$\boxed{\sum_i w_i f_i = 1.} \tag{97}$$

Hence no factorization of the form (96) exists, and orders one and zero are unreachable. This is a no-go for the complete parallel invariant two-wave-factor family, not for mixed-order or enlarged systems, triangular Green extensions, or the equation-cone curvature-to-metric route. Its exact receipt is `curved_endpoint_metric_factorization_no_go.json`; the nonzero multipliers are printed in Appendix C.

The parent-detour route is now complete at the causal-homotopy level. For the flat adjoint-tractor connection, the Yang–Mills detour complex of Ref. [3] has ranks $15 \to 60 \to 60 \to 15$ and a degreewise normally hyperbolic parent witness. Exact Kostant contraction and finite homological perturbation terminate after two terms and compress it to ranks $4 \to 9 \to 9 \to 4$. In curved PBW normal form the former 48-entry commuting-derivative defect decomposes into a complete ten-dimensional $SO(3)$-invariant Hom basis: two time-derivative, five spatial-derivative, and three algebraic channels. The curvature action on the derivative suffix, form slot and graded adjoint-tractor slot reconstructs all ten channels from the Kostant data, with no fitted coefficient, and cancels all 48 entries. The one-form convention is fixed independently by $-R_M^T$ on $\Lambda^1 T^*$ together with $\mathrm{ad}_M = [M, \cdot]$; the other three sign/transpose choices fail the exact harmonic-carrier gate.

The resulting curved differential maps $i_0, i_1, p_0, p_1, h_0, h_1$ are finite-order and support local. Their inclusion, projection, homotopy and cyclic-adjoint defects all vanish. Moreover the complete compressed parent middle operator, not merely its principal symbol, satisfies

$$P_{\mathrm{mid}}^{\mathrm{parent,comp}} = -2S^T J_{\mathrm{met}}\, \bar{B}\, S \tag{98}$$

coefficientwise at derivative orders zero through four. Thus the parent advanced and retarded homotopies transfer directly:

$$\Lambda_{\mathrm{TF},\pm} = p\Lambda_{\mathrm{parent},\pm} i, \qquad q_{\mathrm{TF}}\Lambda_{\mathrm{TF},\pm} + \Lambda_{\mathrm{TF},\pm} q_{\mathrm{TF}} = 1, \qquad \Lambda_{\mathrm{TF},+}^\sharp = \Lambda_{\mathrm{TF},-}. \tag{99}$$

Finite-order pre- and postcomposition preserve the parent causal support.

The four trace/Weyl rows are attached without a nonlocal projector. Put $d = \frac{1}{2}\,\mathrm{div}$ and let the local shear from split to unshifted endpoint coordinates have

$$U_G = \begin{pmatrix} 1 & 0 \\ -d & 1 \end{pmatrix}, \qquad U_I = \begin{pmatrix} 1 & d^\sharp \\ 0 & 1 \end{pmatrix}, \qquad U_M = U_E = 1. \tag{100}$$

Its inverse is obtained by reversing the off-diagonal signs, $U^\sharp = U^{-1}$, and $qU = U(q_{\mathrm{TF}} \oplus q_{\mathrm{tr}})$. The trace complex is two pointwise identity doublets with contraction h_{tr}, so the complete thirty-row endpoint homotopy is

$$\Lambda_{\mathrm{end},\pm} = U(\Lambda_{\mathrm{TF},\pm} \oplus h_{\mathrm{tr}})U^{-1}. \tag{101}$$

Combining this with the certified 356/30 hybrid projector gives the full all-row formula

$$\boxed{\Lambda_{\mathrm{prol},\pm} = H_{\mathrm{alg}} + i_{\mathrm{end}}\Lambda_{\mathrm{end},\pm}p_{\mathrm{end}}}, \qquad Q_{\mathrm{prol}}\Lambda_{\mathrm{prol},\pm} + \Lambda_{\mathrm{prol},\pm}Q_{\mathrm{prol}} = 1. \tag{102}$$

Every algebraic term is finite-order local, hence $\operatorname{supp}\Lambda_{\mathrm{prol},\pm}f \subseteq J^\pm(\operatorname{supp} f)$; cyclicity gives $\Lambda^\sharp_{\mathrm{prol},+} = \Lambda_{\mathrm{prol},-}$. This promotes `causal_green_homotopy`. It does not construct a canonical Green inverse for D_{TF}, a global identity $\Lambda_{\mathrm{end},\pm} = W_0G_{\mathrm{end},\pm}$, or a prolonged witness $QW + WQ$ with degreewise Green inverses. Those stronger objects remain unproved but are unnecessary for the direct transferred homotopy (102). The coefficient and assembly receipts are respectively

`adjoint_tractor_bgg_curved_pbw.json`,
`curved_full_prolonged_green_homotopy_assembly.json`.

The selected causal route therefore has the following explicit flags:

`curved_EB_equations`,	`curved_EB_first_order_closure`,	
`curved_EB_symmetric_hyperbolicity`,	`curved_sourced_constraint_identity`,	
`curved_constraint_propagation`,	`EAL_curvature_spectrum_match`,	
`support_local_prolongation_retract`,	`prolonged_BV_operator_identity`,	(103)
`prolonged_green_witness`,	`curvature_causal_green_operators`,	
`causal_green_homotopy`,	`causal_quasi_isomorphism`,	
`residual_endpoint_recovery`,	`SO42_equivariant_transport`,	
`prolonged_current_comparison`.		

Thirteen flags in (103) are true: the curved equations, first-order closure, symmetric hyperbolicity, sourced identity, constraint propagation, all-level spectrum, support-local prolongation retract, prolonged BV operator identity, off-shell prolonged current comparison, all-row causal Green homotopy, causal quasi-isomorphism, residual endpoint recovery, and equivariant transport. The two false flags are the implementation-specific `prolonged_green_witness` and `curvature_causal_green_operators`. They refer to a canonical D_{TF} same-sided inverse and a degreewise $QW + WQ$ realization, neither of which is asserted or required by the direct tractor transfer. The paper constructs degree-(-1) maps satisfying

$$Q_{\mathrm{prol}}\Lambda_\pm + \Lambda_\pm Q_{\mathrm{prol}} = 1, \qquad \operatorname{supp}\Lambda_\pm f \subseteq J^\pm(\operatorname{supp} f). \tag{104}$$

This is the transferred formula (102), not a Green inverse for the canonical D_{TF} block. Homogeneous constraint propagation and symmetric hyperbolicity by themselves would not have proved (104); the parent Green theorem and the exact curved cyclic retract supply the missing sourced causal operation. Nor does a nonequivariant vector-space identification suffice for the next step. Here the causal difference

$$\Lambda = \Lambda_+ - \Lambda_- : \Gamma_c(\mathcal{C}_{\mathrm{prol}})[1] \longrightarrow \Gamma_{sc}(\mathcal{C}_{\mathrm{prol}}) \tag{105}$$

is a quasi-isomorphism. Its cutoff inverse is represented by $[u] \mapsto [Q(\chi_+ u)]$; the two cohomology-inverse identities follow from the past-compact and future-compact contractions and the exact support sequence. Since S^3 is compact, $\Gamma_{sc}(\mathcal{C}_{\rm prol}) = \Gamma^\infty(\mathcal{C}_{\rm prol})$.

For each of the fifteen conformal-Killing generators ξ_a, the compact source $j_a = Q(\chi\xi_a)$ satisfies $[\Lambda j_a] = [\xi_a]$. The dual quotient sections obey the complementary identity and the endpoint pairing is perfect. Their compact grading is $4_{-1} \oplus 7_0 \oplus 4_{+1}$, the suspension scalar is $+1$, and the curvature mapping-cylinder and auxiliary pairs are contractible; hence they add no second ghost or momentum copy. The causal/Cauchy bridge preserves the residual $SO(4,2)$ action up to a certified chain homotopy. If ρ is any local conformal chain generator and $\kappa = [Q, \chi]$, then

$$[\kappa, \rho] = [Q, [\chi, \rho]]. \tag{106}$$

The homotopy has compact support on every spacelike-compact section because $d\chi$ lies in a compact time slab and S^3 is compact. Together with the natural curvature mapping cylinder, smooth global BGG equivariance, and the all-level $E/A/L$ exhaustion, this induces the strict $SO(4,2)$ action on cohomology. The chosen Cauchy split itself need not be strictly equivariant.

The certified odd cyclic mapping cylinder gives the quadratic parent

$$S_{\rm prol} = \frac{1}{2}\langle \Phi, D\Omega Q_{\rm prol}\Phi\rangle, \qquad I^*\omega_{\rm prol} - \omega_{\rm aux} = d\beta + Q\gamma, \qquad \beta = \delta Y_I, \quad \gamma = 0, \tag{107}$$

where Y_I is the finite Lagrange concomitant of the T and A shears. The current certificate now binds the corrected core-chain digest and checks this pullback explicitly. Hence the flag

`prolonged_current_comparison`

is true. Moreover $\Lambda_+^\sharp = \Lambda_-$ makes $\Delta_\Lambda = \Lambda_+ - \Lambda_-$ graded antisymmetric. The common algebraic and trace contractions cancel in Δ_Λ, and cyclic transfer reduces its pairing to the parent Green/current identity. The two $d + Q$ current improvements then identify the prolonged, auxiliary, metric, Cauchy and causal pairings on cohomology. The normalized result is $+I_E \oplus (-I_A) \oplus (-I_L)$ before residual reduction.

Theorem 2.9 (Covariant causal transport)**.** *For the selected closed-cylinder BV–BFV polarization, the chain (72) consists of pairing-compatible quasi-isomorphisms. It recovers exactly the fifteen conformal-Killing ghost classes and their fifteen dual endpoint classes, introduces no duplicate residual sector, and is $SO(4,2)$-equivariant on cohomology. Consequently*

$$\boxed{H^4_{\rm cov} = \operatorname{span}\{[W_+^2], [W_-^2]\}, \qquad \operatorname{sig} G_{\rm cov} = (2,0), \qquad G_{\rm cov} = I_2.} \tag{108}$$

These are residual vertex/deformation classes. Their Hermitian form is positive-definite; I_2 is its value after the certified basis, orientation, and unit-overlap normalization. The theorem transports the certified residual result; it does not recompute Chevalley–Eilenberg cohomology in auxiliary variables or assert a positive graviton Fock norm.

Proof. The support exact sequence turns the same-sided homotopies in (104) into the map (105), which is a quasi-isomorphism. Compactness of S^3 identifies its target with all smooth global solutions. The cutoff identities above recover the fifteen endpoint pairs, while the support-local SDRs contract every auxiliary and curvature copy. Equation (106) gives homotopy-equivariant residual transport. Cyclicity and the two current improvements identify the causal pairing with the normalized energy and residual pairings. Theorem 6.2 then gives (108). ☐

The receipts are

`curved_causal_transport_recognition.json`,
`curved_SO42_causal_transport_recognition.json`,
`curved_direct_causal_pairing_transport.json`,
`covariant_H4_transport.json`, `covariant_gram_transport.json`.

The known obstruction to factorizing the gauge-invariant spin-two conformal operator away from Einstein backgrounds is relevant context [17], but is not used as the no-go theorem (60).

3 The on-shell Weyl module

3.1 Character and cylinder towers

We now define the coefficient module used in the exact residual calculation. A positive-chirality Weyl curvature is a conformal primary with labels

$$(\Delta; j_L, j_R) = (2; 2, 0), \tag{109}$$

and the opposite chirality is its parity conjugate. The Bach equation is in $(4; 1, 1)$ and its differential identity is in $(5; \frac{1}{2}, \frac{1}{2})$. The resulting one-chirality character is

$$\chi_+(q) = \chi_{(2;2,0)} - \chi_{(4;1,1)} + \chi_{(5;\frac{1}{2},\frac{1}{2})} = \frac{5q^2 - 9q^4 + 4q^5}{(1-q)^4}. \tag{110}$$

This defines the abstract algebraic positive-energy module $\mathcal{W}_+$. The character and tower calculation below, together with the explicit intertwiner and metric preimages of Theorem 2.5, identify it with the geometric chiral system.

On $\mathbb{R} \times S^3$, $\mathcal{W}_+$ decomposes into three multiplicity-free towers:

$$E_n^+ : \quad \left(\frac{n+2}{2}, \frac{n-2}{2}\right), \qquad \dim E_n = (n+3)(n-1), \qquad n \geq 2, \tag{111}$$

$$A_n^+ : \quad \left(\frac{n}{2}, \frac{n-2}{2}\right), \qquad \dim A_n = (n+1)(n-1), \qquad n \geq 3, \tag{112}$$

$$L_n^+ : \quad \left(\frac{n}{2}, \frac{n-4}{2}\right), \qquad \dim L_n = (n+1)(n-3), \qquad n \geq 4. \tag{113}$$

The negative-chirality towers exchange j_L and j_R. Summing (111)–(113) reproduces (110). For both chiralities the first levels have dimensions

$$\dim \mathcal{H}_2 = 10, \quad \dim \mathcal{H}_3 = 40, \quad \dim \mathcal{H}_4 = 82, \quad \dim \mathcal{H}_5 = 136, \quad \dim \mathcal{H}_6 = 202. \tag{114}$$

The labels E, A, L are cylinder oscillator labels only. In particular, they should not be interpreted as three independently signable conformal representations. Proper conformal transformations mix the towers.

3.2 Invariant form and exact generator action

In normalized cylinder coordinates the unique standard-real, parity-compatible invariant form is, up to overall convention,

$$J_{\text{conf}} = +I_E \oplus (-I_A) \oplus (-I_L). \tag{115}$$

It is nondegenerate and indefinite. The proper conformal lowering and raising generators satisfy

$$(K_q^-)^\sharp = K_q^+, \qquad X^\sharp = X \quad (X = D, SU(2)_L, SU(2)_R), \tag{116}$$

where $X^\sharp = J_{\text{conf}}^{-1} X^\dagger J_{\text{conf}}$.

Multiplicity one reduces each allowed lowering block to a scalar reduced coefficient. With source energy n, the six stable families are

$$\begin{aligned}
c_{EE}(n) &= \sqrt{\frac{2(n-1)(n+1)(n+3)}{n+2}}, & c_{AE}(n) &= \sqrt{\frac{8(n-1)}{(n-2)(n+2)}}, \\
c_{AA}(n) &= \sqrt{\frac{2(n-3)(n-1)(n+2)}{n-2}}, & c_{LE}(n) &= -\sqrt{\frac{2(n-3)}{n-2}}, \\
c_{LA}(n) &= \sqrt{\frac{8}{n-2}}, & c_{LL}(n) &= \sqrt{2(n-2)(n+1)}.
\end{aligned} \tag{117}$$

Together with Clebsch–Gordan coefficients and (116), these determine the all-level one-particle $\mathfrak{so}(4,2)$ action. The exact reconstruction verifies the conformal commutators on every interior level of a finite energy buffer; the top shell is used only as a buffer and is never mistaken for a finite conformal representation.

The free Fock coefficient module is the bosonic second quantization

$$\mathcal{F}_{\mathcal{W}} = \text{Sym}(\mathcal{W}_+ \oplus \mathcal{W}_-), \tag{118}$$

with multiplicative lift of (115). The residual generators preserve oscillator particle number.

4 Hamiltonian moment-map normalization

The residual action is not introduced as an arbitrary matrix representation. In oscillator coordinates z, write the symplectic form as

$$\Omega = i\, d\bar{z}\, J \wedge dz. \tag{119}$$

If a conformal generator X acts linearly by $\delta_X z = K_X z$, its quadratic Hamiltonian moment map is

$$\mu_X(z, \bar{z}) = \bar{z}\, M_X z, \qquad M_X = J K_X, \qquad d\mu_X = \iota_{X_X} \Omega. \tag{120}$$

The J-adjoint identities ensure the correct reality of μ_X.

4.1 The conformal Taub constraint

The same residual parameters arise as reducibilities of the local Diff×Weyl complex. Let

$$\epsilon = (\xi, \sigma), \qquad K(\xi, \sigma) = 0, \qquad \sigma = -\frac{1}{4} \bar{\nabla}_\mu \xi^\mu, \tag{121}$$

and let h satisfy $B^{(1)}[h] = 0$. With the Taylor convention

$$g(\lambda) = \bar{g} + \lambda h + \frac{\lambda^2}{2} k + O(\lambda^3), \tag{122}$$

the second-order equation is

$$B^{(1)}[k] + B^{(2)}[h,h] = 0. \tag{123}$$

Proposition 4.1 (Conformal Taub solvability constraint)**.** *For every residual conformal reducibility ϵ, the current*

$$J^\mu_\epsilon[h] := \xi_\nu B^{(2)\mu\nu}[h,h] \tag{124}$$

is conserved. If h is tangent to a second-order family of Bach-flat metrics on the compact cylinder, then

$$\boxed{\mathcal{T}_\epsilon[h] := \int_{S^3} n_\mu \xi_\nu B^{(2)\mu\nu}[h,h], d\Sigma = 0.} \tag{125}$$

Proof. At second order around a Bach-flat background, the exact divergence and trace identities imply $\bar{\nabla}_\mu B^{(2)\mu\nu}[h,h] = 0$ and $B^{(2)\mu}{}_\mu[h,h] = 0$ whenever $B^{(1)}[h] = 0$. Hence

$$\bar{\nabla}_\mu J^\mu_\epsilon = B^{(2)\mu\nu} \bar{\nabla}_{(\mu}\xi_{\nu)} = -\sigma B^{(2)\mu}{}_\mu = 0. \tag{126}$$

If (123) has a solution, the charge equals minus the integral of $\xi_\nu B^{(1)\mu\nu}[k]$. For a reducibility parameter, the linearized Noether current is a spatial divergence. Since $\partial S^3 = \varnothing$, its integral vanishes. This is the standard compact-Cauchy linearization-stability argument [29, 30]. □

The covariant phase-space Noether identity identifies this same quadratic constraint with the Hamiltonian of the residual linear action:

$$\mu_\epsilon(h) = \frac{1}{2}\Omega_\Sigma(h, R_\epsilon h) = \mathcal{T}_\epsilon[h] \tag{127}$$

in the Taylor and orientation conventions fixed above. Changing the polarization convention moves the corresponding factor of two between the bilinear kernel and its quadratic polynomial. Both sides form equivariant adjoint/coadjoint $\mathfrak{so}(4,2)$ modules. Since the complexified adjoint module is simple, a nonzero equivariant comparison is unique up to one scalar. This is also the local-BRST relation between reducibilities and conserved-current classes [20, 19]; the covariant Hamiltonian identity is the higher-derivative counterpart of the standard Noether-charge construction [25].

The equality in (127) has an abstract Hessian proof. Let $H_\epsilon[g]$ be the covariant constraint Hamiltonian. Because ϵ fixes the background, $H_\epsilon[\bar{g}] = dH_\epsilon|_{\bar{g}} = 0$. Its Hessian on linearized solutions is simultaneously

$$d^2 H_\epsilon|_{\bar{g}}(h,h) = \Omega_\Sigma(h, R_\epsilon h) = 2\mathcal{T}_\epsilon[h]. \tag{128}$$

The first equality is the Hamiltonian definition of the residual linear action; the second is the quadratic term in the same nonlinear constraint with the Taylor convention of (123). Thus the direct component calculations below fix conventions and the common scalar rather than supplying fifteen logically independent proofs.

Computational Proposition 4.2 (All-energy Taub/moment-map normalization). *For $S_{\mathrm{red}} = -S_{\mathrm{HH}}/2$ modulo Euler, the canonical fifteen-component quadratic kernel is*

$$M_X^{\mathrm{can}} = -\frac{1}{2} J K_X. \tag{129}$$

The independent quadratic Bach-source calculation uses a different normalization for the cylinder waves and conformal-Killing vectors. In that convention the proper-conformal kernel is

$$M_{\mathrm{Taub}} = -\frac{\sqrt{2}}{4\pi} J K^{-}. \tag{130}$$

All six stable lowering families are therefore fixed at arbitrary source energy by (117). The D component agrees directly with the quadratic Ostrogradsky Noether Hamiltonian. Two independent integrations of $n_\mu \xi_\nu B^{(2)\mu\nu}[h_1, h_2]$ reproduce the $A_3 \to E_2$ and $L_4 \to A_3$ reduced coefficients. Exact conformal equivariance then fixes the remaining magnetic components and compact charges.

The proof is constructive. The all-level matrices obey the complete interior conformal algebra and the J-adjoint identities. Multiplication by $-J/2$ therefore gives the canonical equivariant moment map. Direct evaluation of the fourth-order and vector quadratic Hamiltonians gives $+\omega$, $-\omega$, and $-\omega$ on the normalized E, A, L oscillators for S_{HH}, and hence the displayed D kernel for S_{red}. The two curvature-derived proper-conformal components fix the single conversion

$$M_{K^-}^{\mathrm{raw}} = \frac{\sqrt{2}}{2\pi} M_{K^-}^{\mathrm{can}}. \tag{131}$$

The exact all-n formulas and their energy-six regression buffer are generated by `symbolic/verify_conformal_taub_moment_map_all_levels.py`.

After Proposition 8.5, the quadratic Bach source also defines a canonical Kuranishi class

$$\kappa_2(h, h) = [B^{(2)}[h, h]] \in \operatorname{coker} K^{\sharp}. \tag{132}$$

Its image under endpoint duality evaluates on a reducibility as

$$\Theta(\kappa_2(h, h))(\epsilon) = \langle \epsilon, B^{(2)}[h, h] \rangle = \mathcal{T}_\epsilon[h] = \mu_\epsilon(h). \tag{133}$$

The exact quotient invariance, direct D normalization, two independent Bach-source seeds, and equivariant fifteen-component completion are composed by `symbolic/verify_conformal_taub_obstruction_map.py`. This does not compute the separate time-slice transgression from the bulk endpoint degree to BFV ghost momentum.

This section supplies the Hamiltonian normalization of the residual action and its Taub interpretation. It reconstructs all components by equivariance after two direct Bach-source normalizations; it does not claim that every magnetic block was redundantly recomputed as a separate curvature integral. The cohomology theorem below depends only on the exact module action.

5 Residual BRST and Cartan homotopy reduction

5.1 The absolute residual complex

Let

$$\mathfrak{g} = \mathfrak{so}(4, 2) \tag{134}$$

with compact grading

$$\mathfrak{g} = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_{+1}, \qquad \dim(\mathfrak{g}_{-1}, \mathfrak{g}_0, \mathfrak{g}_{+1}) = (4, 7, 4). \tag{135}$$

Here D and $\mathfrak{so}(4)$ lie in $\mathfrak{g}_0$, while proper conformal lowering and raising generators lie in $\mathfrak{g}_{-1}$ and $\mathfrak{g}_{+1}$. The dual ghosts carry the opposite compact degree.

For a coefficient module $V \subset \mathcal{F}_{\mathcal{W}}$, the absolute residual complex is

$$C^q(V) = V \otimes \Lambda^q \mathfrak{g}^*, \tag{136}$$

with Chevalley–Eilenberg differential

$$d = c^a \rho(G_a) - \frac{1}{2} f^a{}_{bc} c^b c^c \iota_a. \tag{137}$$

It raises ghost number by one and preserves the total compact degree

$$\delta = D_{\mathrm{matter}} + D_{\mathrm{ghost}}. \tag{138}$$

The standard centered polarization uses the product v of the four ghosts dual to the raising generators. It has

$$\mathrm{gh}(v) = 4, \qquad D_{\mathrm{ghost}}(v) = -4. \tag{139}$$

This is the residual ghost convention used in the cylinder construction of [13]. Its compact degree and uniqueness as the bottom rotational scalar follow from the $(4, 7, 4)$ exterior-algebra polarization, and Lemma 6.1 shows that its complementary-degree pairing is canonical up to one overall volume. The selected closed-universe BFV certificate and raw cyclic form fix that volume in the algebraic cylinder model. Theorem 8.12 identifies it with the pairing of the selected algebraic gauge-fixed BV–BFV state domain; the broader Riegert sector of [13] is not included.

5.2 Full residual gauging is a physical assumption

Hamada's background-independence programme identifies the transformations which preserve the cylinder gauge with residual diffeomorphisms and imposes the resulting conformal charges through BRST [12, 13]. This supplies a direct precedent for the complex (136). It does not prove that every quantization of strict pure-Weyl gravity must use the same quotient.

There are at least two distinct cylinder problems.

1. In the *fully gauged residual problem*, the cylinder is a gauge representative of a closed generally covariant system. All fifteen conformal-Killing transformations, including compact time translation, are generated by residual constraints. The ghost for D exists and the Cartan homotopy below is an allowed operation.

2. In the *fixed-background cylinder problem*, D is the Hamiltonian and $SO(4, 2)$ is a global symmetry acting on states or observables. Its eigenspaces are not quotiented. The D ghost and its contraction cannot be used to erase nonzero energy.

Proposition 4.1 gives the first choice a direct linearization-stability interpretation. Within the closed-universe Dirac boundary problem, the slice S^3 has no spatial boundary supporting an independent asymptotic charge, and the residual conformal Killing parameters instead generate quadratic integrability constraints, including the D constraint. Subject to the all-level normalization in (127), full residual gauging is therefore a coherent zero-boundary-charge sector of pure conformal gravity; it is not a device introduced merely to erase positive cylinder energy. Compactness and the Taub identity support this interpretation but do not make it the unique physical quantization.

Adding a boundary or asymptotic region, coupling an external clock, or deparametrizing the cylinder can convert D into a physical global charge. That is the second problem above, not a contradiction of the first. Because $[K^-, K^+]$ contains D, the second problem is not obtained by simply deleting one ghost while leaving the rest of (137) unchanged. It requires a different relative or boundary-charge complex. Theorem 6.2 answers only the first problem. A strict pure-Weyl derivation must state the temporal boundary conditions, the residual gauge group, and the treatment of its charges before calling the result a physical state space.

The precise derived-reduction statement, including the endpoint-duality and BV–BFV suspension proof, is Theorem 8.13 after the local-to-residual transfer has been constructed. Its accompanying remark records the fixed-cylinder and relative alternatives.

5.3 Cartan homotopy

Let ι_D be contraction with the generator D in the residual ghost exterior algebra. The usual Cartan identity is

$$d\iota_D + \iota_D d = \mathcal{L}_D. \tag{140}$$

Theorem 5.1 (Cartan homotopy reduction to compact degree zero)**.** *Decompose the residual complex into total compact-degree eigenspaces,*

$$C = \bigoplus_{\delta} C_\delta, \qquad \mathcal{L}_D|_{C_\delta} = \delta\,\mathrm{id}. \tag{141}$$

Every C_δ with $\delta \neq 0$ is contractible. Hence inclusion of the centered subcomplex is a quasi-isomorphism:

$$\boxed{H^\bullet(C, d) \cong H^\bullet(C_{\delta=0}, d).} \tag{142}$$

Proof. On C_δ with $\delta \neq 0$, define

$$h_\delta = \frac{\iota_D}{\delta}. \tag{143}$$

Equation (140) gives

$$dh_\delta + h_\delta d = \mathrm{id}. \tag{144}$$

Thus every closed vector of nonzero total degree has the explicit primitive $\delta^{-1}\iota_D\Psi$. If $\mathcal{L}_D$ has a nilpotent part, the same proof works wherever $\mathcal{L}_D$ is invertible by using $h = \iota_D\mathcal{L}_D^{-1}$. □

The theorem depends on treating D as a residual gauge generator. If cylinder time translation is retained as a physical Hamiltonian or a boundary charge, contraction by its ghost is not an allowed quotient and (142) does not apply.

5.4 Finite centered inventory

At ghost number four, the minimal residual exterior algebra contains at most the four negative-degree ghosts, so

$$D_{\text{ghost}} \geq -4. \tag{145}$$

At $\delta = 0$ this implies

$$D_{\text{matter}} \leq 4. \tag{146}$$

Every nonvacuum Weyl quantum has weight at least two. Therefore only particle numbers $N = 0, 1, 2$ can contribute to the centered ghost degree selected by the Hamada polarization. All $N \geq 3$ sectors have matter weight at least six and are contractible before any matrix-rank computation.

6 Complete centered cohomology

6.1 The relative weight-four kernel

The complete matter Fock space at weight four is

$$\mathcal{F}_4 = \mathcal{H}_4 \oplus \mathrm{Sym}^2 \mathcal{H}_2, \qquad \dim \mathcal{F}_4 = 82 + 55 = 137. \tag{147}$$

The induced matter form has signature

$$\mathrm{sig}(J_{\mathcal{F}_4}) = (97, 40). \tag{148}$$

The relative conformal-primary conditions are

$$(D - 4)|\Psi\rangle = 0, \qquad R|\Psi\rangle = 0, \qquad K_q^-|\Psi\rangle = 0. \tag{149}$$

Their exact kernel on the full space (147) is two-dimensional. In a spin-two magnetic basis its normalized chiral generators are

$$|W_\pm^2\rangle = \sqrt{\frac{2}{5}}|2, -2\rangle_\pm - \sqrt{\frac{2}{5}}|1, -1\rangle_\pm + \frac{1}{\sqrt{5}}|0, 0\rangle_\pm. \tag{150}$$

They obey

$$\langle W_r^2|J_{\mathcal{F}}|W_s^2\rangle = \delta_{rs}. \tag{151}$$

This relative computation is not by itself an absolute cohomology theorem; incoming exact states must also be excluded.

There is a simple representation-theoretic reason for the number two. The weight-two one-particle space is

$$E_2^+ \oplus E_2^- = (2, 0) \oplus (0, 2). \tag{152}$$

The symmetric square decomposes as

$$\mathrm{Sym}^2(2, 0) \oplus \big[(2, 0) \otimes (0, 2)\big] \oplus \mathrm{Sym}^2(0, 2). \tag{153}$$

There is one compact scalar in each same-chirality symmetric square and no scalar in the mixed product. Since $E_2^\pm$ are lowest-weight spaces, every K_q^- annihilates both factors, so these two scalar singlets are relative primaries. By contrast, the one-particle weight-four space contains

$$(3, 1) \oplus (2, 1) \oplus (2, 0) \tag{154}$$

and its parity conjugate, with no compact scalar. Thus representation theory predicts the two relative candidates before the matrix calculation. The absolute computation below is still necessary to exclude exactness and to prove the one-particle acyclicity. A complete Hochschild–Serre or Kostant-style derivation of the latter ranks would be a useful conceptual replacement for the large matrices [7].

6.2 Absolute kernel and image

The cutoff-complete absolute calculation supplies that missing test. For one chirality in the one-particle coefficient module, the centered complex around ghost number four has exact dimensions

$$C_0^3(290) \xrightarrow{d_3} C_0^4(1311) \xrightarrow{d_4} C_0^5(3657). \tag{155}$$

The entries lie in $\mathbb{Q}(i, \sqrt{2}, \sqrt{3}, \sqrt{5})$. Good-prime reduction modulo 241, combined with exact nilpotency, fixes the characteristic-zero ranks to

$$\operatorname{rank} d_3 = 260, \qquad \operatorname{rank} d_4 = 1051. \tag{156}$$

Since the ranks sum to 1311,

$$H^4_{\delta=0,N=1} = 0. \tag{157}$$

Parity supplies the identical result for the opposite chirality.

The lowest two-particle block begins at matter weight four. Exterior saturation makes its beginning

$$0 \longrightarrow C_0^4(55) \xrightarrow{d_4} C_0^5(385) \xrightarrow{d_5} C_0^6(1155). \tag{158}$$

There is no incoming C_0^3 space: three residual ghosts have compact degree at least -3 and cannot center a two-particle coefficient of minimum weight four. Exact nilpotency gives $d_5 d_4 = 0$. The two vectors (150) lie in $\ker d_4$, while a good-prime minor gives $\operatorname{rank} d_4 \geq 53$. The displayed independent kernel vectors give $\operatorname{rank} d_4 \leq 53$. Therefore

$$H^4_{\delta=0,N=2} = \operatorname{span}\{[W_+^2], [W_-^2]\}. \tag{159}$$

There is also a deliberately independent small cross-check of this central rank. The standard-library-only executable `symbolic/verify_conformal_residual_rank53_independent.py` reconstructs the integral spin-two action on $\mathrm{Sym}^2((2,0) \oplus (0,2))$ without importing the conformal generator, Fock or BRST implementations used above. It finds the block ranks $14+25+14 = 53$ at two good primes and exhibits one exact singlet in each chirality, so the exact characteristic-zero rank is 53. This check does not replace the one-particle absolute window, full CE nilpotency, BV–BFV transfer or covariant audits.

For the vacuum coefficient module the complex is ordinary Lie-algebra cohomology. Since

$$H^\bullet(\mathfrak{so}(4,2); \mathbb{C}) \cong \Lambda(u_3, u_5, u_7), \tag{160}$$

one has

$$H^4_{\delta=0,N=0} = 0. \tag{161}$$

6.3 The canonical residual CE pairing

The residual ghost normalization is fixed almost completely by the compact grading. Write

$$\mathfrak{g} = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_{+1}, \qquad \dim(\mathfrak{g}_{-1}, \mathfrak{g}_0, \mathfrak{g}_{+1}) = (4, 7, 4), \tag{162}$$

and choose nonzero determinant vectors

$$v_- \in \det(\mathfrak{g}_{+1}^*), \qquad \Theta_0 \in \det(\mathfrak{g}_0^*), \qquad v_+ \in \det(\mathfrak{g}_{-1}^*), \tag{163}$$

with $v_+ = v_-^\dagger$. Their product

$$\Omega_{\mathfrak{g}} = v_+ \wedge \Theta_0 \wedge v_- \tag{164}$$

is the top Berezin volume on all fifteen residual ghosts. Define the polarized pairing by coefficient extraction,

$$(\alpha, \beta)_{\mathrm{gh}} := [\alpha^\dagger \wedge \Theta_0 \wedge \beta]_{\Omega_{\mathfrak{g}}}. \tag{165}$$

Lemma 6.1 (Canonical centered ghost pairing)**.** *After fixing the orientation and the overall residual volume,* (165) *is the unique determinant-saturating pairing compatible with the* $(4, 7, 4)$ *polarization. It obeys*

$$(v_-, v_-)_{\mathrm{gh}} = 1. \tag{166}$$

Because $\mathfrak{so}(4, 2)$ *is semisimple and hence unimodular, the Chevalley–Eilenberg differential is graded skew for the associated complementary-degree Poincaré pairing.*

Proof. Each of the three determinant lines is one-dimensional. Once their product is oriented and normalized, coefficient extraction gives (166) and leaves no matrix freedom. Unimodularity makes the top CE volume invariant, so integration by parts in the exterior algebra gives graded skew-adjointness of the CE differential. □

Consequently exact cochains are orthogonal to closed complementary-degree cochains, and the pairing descends to residual CE cohomology.

In Hamada's oscillator notation, v_- is the product of the four proper-conformal ghosts, Θ_0 contains the dilation/time ghost and the six rotation ghosts, and the bra supplies the four conjugate ghosts [13]. The factor of i is a real-structure convention. The repository exterior-algebra certificate verifies the saturation, Hermiticity, compact degree -4, and unit overlap directly.

The fixed insertion is not a nondegenerate inner product on every individual ghost-number block. It is the polarized form of the canonical complementary-degree CE pairing. It is sufficient here because the incoming two-particle space is empty and the two normalized representatives are fixed. Multiplying (166) by the matter restriction (151) gives their exact residual cohomological Gram matrix. It is Hermitian and sesquilinear on the complex oscillator module; the parity-compatible real structure gives its real restriction. This is an intrinsic statement about the chosen residual CE complex. We reserve the phrase "transferred pure-Weyl BV cohomological pairing" for the cyclic-transfer statement in Section 8. The selected algebraic BV–BFV reduction induces this residual orientation and overall volume by Theorem 8.12.

Theorem 6.2 (Complete cohomology under full residual conformal gauging)**.** *Let the coefficient module be the polarized algebraic free symmetric-algebra (Fock) realization* $\mathcal{F}(\mathcal{W}_+ \oplus \mathcal{W}_-)$ *of the linearized classical theory, let the residual algebra be* $\mathfrak{so}(4, 2)$*, treat* D *as a gauge generator, and use the centered ghost polarization* (139)*. Then the complete minimal absolute residual cohomology at the selected ghost number four is*

$$\boxed{H^4_{\mathrm{res,min}} = \mathrm{span}\{[W_+^2], [W_-^2]\}, \qquad \mathrm{sig}\, G_{\mathrm{res}} = (2, 0), \qquad G_{\mathrm{res}} = I_2.} \tag{167}$$

The two survivors are residual vertex/deformation classes. Their Hermitian form is positive-definite, and the displayed I_2 *uses the fixed chiral basis, ghost orientation, and unit-overlap normalization. There is no one-particle class and no higher-matter-weight class in this minimal centered polarization.*

Proof. The Cartan homotopy, Theorem 5.1, eliminates every nonzero total compact degree. The ghost-degree floor then reduces the centered inventory to $N = 0, 1, 2$. Equations (161), (157), and (159) exhaust those sectors. Equations (151) and (166) give the pairing. □

Remark 6.3 (What is positive). The theorem does not turn the indefinite one-particle module (115) into a positive particle space. It removes the one-particle sector from this absolute residual cohomology. The positive-definite $(2,0)$ form, normalized to I_2, belongs to two ghost-dressed scalar vertex/deformation classes.

7 Descent and the dynamical/topological split

7.1 Why weight four is selected

Let V_h be a scalar conformal primary of weight h, and let

$$\omega = \frac{1}{4!}\epsilon_{\mu\nu\rho\sigma}c^\mu c^\nu c^\rho c^\sigma \tag{168}$$

be the top lowering-ghost product. In the residual descent conventions,

$$sV_h = c^\mu \nabla_\mu V_h + \frac{h}{4}(\nabla_\mu c^\mu)V_h, \qquad s\omega = -\omega\nabla_\mu c^\mu, \qquad \omega c^\mu = 0. \tag{169}$$

It follows that both the integrated density and the unintegrated ghost-dressed operator have the same closure defect, proportional to $h/4 - 1$. Hence

$$\boxed{[\omega V_4] \longleftrightarrow \left[\int_M d^4x\,\sqrt{-g}\,V_4\right].} \tag{170}$$

The -4 compact degree of the ghost factor is therefore the conformal counterpart of the four-dimensional volume form, rather than an accidental normal-ordering shift. This weight-four state–operator mechanism and a Weyl-square example were already explicit in [13]; the new use here is the absolute exhaustion and pairing calculation of Section 6.

Three cohomological quotients appear and must not be identified without the transfer theorem:

$$H^4\big(\mathfrak{so}(4,2);\mathcal{F}_\mathcal{W}\big), \qquad H^{0,4}_{\text{loc}}(s \mid d), \qquad \frac{\{\text{integrated local functionals}\}}{\{\text{allowed boundary functionals}\}}. \tag{171}$$

The first is the residual Chevalley–Eilenberg group computed exactly in this paper. The second is local BRST cohomology modulo spacetime descent [19]. The third depends on topology and boundary conditions. Equation (170) identifies representatives in the residual construction; its promotion to an isomorphism with the full local BRST group modulo all allowed spacetime boundary functionals is an additional local-descent question, not part of the polarized state theorem.

7.2 Parity basis

Parity exchanges W_+^2 and W_-^2. Define

$$e = \frac{W_+^2 + W_-^2}{\sqrt{2}}, \qquad o = \frac{W_+^2 - W_-^2}{\sqrt{2}}. \tag{172}$$

Up to the orientation-dependent factor of i in Lorentzian self-duality conventions, descent identifies

$$e \sim C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}, \qquad o \sim C_{\mu\nu\rho\sigma}\widetilde{C}^{\mu\nu\rho\sigma}. \tag{173}$$

Because (172) is orthogonal, its Gram matrix remains I_2.

7.3 Euler–Lagrange quotient

Let

$$\mathcal{E} : H^4_{\text{res,min}} \longrightarrow \{\text{local equation and gauge deformations}\} \tag{174}$$

be the Euler–Lagrange map obtained by varying the integrated density. In the conventions of Section 2,

$$\mathcal{E}(e) = \lambda_B B_{\mu\nu}, \qquad \lambda_B \neq 0, \qquad \mathcal{E}(o) = 0. \tag{175}$$

The first identity is the Weyl-square variation. Under locality, smoothness, Lorentz invariance, and a four-derivative bound, it is the unique nontrivial self-interaction of a single linear conformal graviton modulo equivalence [18].

The second identity is Chern–Weil transgression. In a local frame,

$$P_4 = \operatorname{Tr}(R \wedge R) = dQ_3(\Gamma), \tag{176}$$

where

$$Q_3(\Gamma) = \operatorname{Tr}\left(\Gamma \wedge d\Gamma + \frac{2}{3}\Gamma \wedge \Gamma \wedge \Gamma\right). \tag{177}$$

Its variation is

$$\delta P_4 = 2d\operatorname{Tr}(\delta\Gamma \wedge R), \tag{178}$$

using the Bianchi identity. On a finite cylinder, in a chosen global frame,

$$\int_{[t_1,t_2]\times S^3} P_4 = \mathrm{CS}[\Gamma(t_2)] - \mathrm{CS}[\Gamma(t_1)]. \tag{179}$$

Thus fixed-endpoint or compactly supported variations have no bulk Euler–Lagrange derivative.

Equations (175)–(179) give the exact sequence

$$0 \longrightarrow \operatorname{span}\{o\} \longrightarrow H^4_{\text{res,min}} \xrightarrow{\mathcal{E}} \operatorname{span}\{B_{\mu\nu}\} \longrightarrow 0. \tag{180}$$

Quotienting variationally trivial local bulk functionals yields

$$H^4_{\text{dyn}} = H^4_{\text{res,min}}/\ker\mathcal{E} \cong \operatorname{span}\{[C^2]\}, \qquad \boxed{G_{\text{dyn,res}} = I_1.} \tag{181}$$

Equivalently,

$$\boxed{I_2 = I_1^{\text{dynamical}} \oplus I_1^{\text{topological}}.} \tag{182}$$

The odd class is only locally and variationally trivial. A gravitational theta parameter can affect boundaries, horizons, contact terms, large frame transformations, and nontrivial topology [32]. We do not quotient those global data without specifying the boundary problem.

8 The local-to-residual transfer

Theorem 6.2 begins with the locally reduced Weyl module. This section states exactly what is needed to identify it with the complete free pure-Weyl BV answer.

8.1 The free Fock lift is formal

Proposition 8.1 (Algebraic symmetric-algebra lift)**.** *Let (C, q) be the algebraic one-particle free complex over characteristic zero, with q extended as a derivation to* $\operatorname{Sym} C$*. Then*

$$\boxed{H(\operatorname{Sym} C, q) \cong \operatorname{Sym} H(C, q).} \tag{183}$$

In particular, once the one-particle coefficient module is $\mathcal{W}_+ \oplus \mathcal{W}_-$, its free many-particle cohomology is $\mathcal{F}(\mathcal{W}_+ \oplus \mathcal{W}_-)$.

Proof. Degree by degree, $C^{\otimes N}$ obeys the Künneth theorem. Taking symmetric coinvariants under the finite group S_N is exact in characteristic zero, so $H(\operatorname{Sym}^N C, q) \cong \operatorname{Sym}^N H(C, q)$. Taking the algebraic direct sum over N proves (183). □

Thus the Fock-space coefficient module is not an independent field-theory input and is not the unpolarized classical BV complex. Proposition 8.1 is the symmetric-algebra lift of the free linearized BRST differential. Its selected Hilbert/Krein symmetric-Fock realization and the closed residual operator are supplied separately by Theorem 2.6; that theorem does not claim an arbitrary topological tensor-product completion.

8.2 Equivariant Cartan transfer

Let (C, q) be the local coefficient complex after the fifteen conformal-Killing zero modes have been separated, and let $H = H(q)$. In the field-theoretic application to states, C denotes the time-slice-reduced, polarized complex of (223), not the entire unpolarized bulk BV tangent complex. Suppose there is a strong deformation retract

$$(H, 0) \underset{p}{\overset{\jmath}{\rightleftarrows}} (C, q), \qquad \jmath p = \mathrm{id} - qs - sq, \qquad p\jmath = \mathrm{id}_H. \tag{184}$$

Let Δ contain the residual zero-mode differential and mixed terms, so $Q = q + \Delta$. With the sign convention in (184), homological perturbation gives

$$I = (\mathrm{id} + s\Delta)^{-1}\jmath, \qquad P = p(\mathrm{id} + \Delta s)^{-1}, \qquad Q_H = p\Delta(\mathrm{id} + s\Delta)^{-1}\jmath. \tag{185}$$

If

$$[D, q] = [D, \Delta] = [D, s] = 0, \qquad pD = D_H p, \qquad D\jmath = \jmath D_H, \tag{186}$$

then the transferred operators

$$\iota_D^H = P\iota_D I, \qquad \mathcal{L}_D^H = P\mathcal{L}_D I \tag{187}$$

obey

$$[Q_H, \iota_D^H]_+ = \mathcal{L}_D^H. \tag{188}$$

Thus every transferred subspace on which $\mathcal{L}_D^H$ is invertible is again contractible.

8.3 Cyclic transfer and the pairing

Let $\sharp$ be the graded adjoint for the full BV/Krein pairing. Suppose the unperturbed retract is cyclic,

$$\jmath^\sharp = p, \qquad (s\Delta)^\sharp = \Delta s. \tag{189}$$

Then taking the adjoint of (185) gives

$$I^\sharp = P. \tag{190}$$

Since $PI = \mathrm{id}_H$,

$$\boxed{I^\sharp I = \mathrm{id}_H.} \tag{191}$$

The dressed inclusion is therefore an exact isometry: contractible local dressing changes representatives without changing the reduced Gram matrix.

Proposition 8.2 (Blockwise cyclic retract)**.** *Let a finite compact-energy and $SO(4)$ block carry a nondegenerate graded pairing, let $q^\sharp = -q$, and suppose the induced pairing on $H(q)$ is nondegenerate. Then the block admits a cyclic strong deformation retract. The algebraic direct sum of these retracts may be chosen compact-degree and $SO(4)$ equivariant.*

Proof. Write $B = \operatorname{im} q$ and choose a nondegenerate representative space $\mathcal{H} \subset \ker q$. In $\mathcal{H}^\perp$, choose an isotropic complement L to B. Then $q : L \to B$ is an isomorphism. Define $s|_B = (q|_L)^{-1}$ and set $s = 0$ on $\mathcal{H} \oplus L$. The decomposition $C = \mathcal{H} \oplus B \oplus L$ gives $qs + sq = 1 - \jmath p$, while the hyperbolic pairing of B and L gives the cyclic adjoint relation. Choosing the splitting in each finite symmetry block and taking the algebraic direct sum preserves compact degree and $SO(4)$ type. □

This proposition does not provide one splitting which is simultaneously equivariant under the noncompact generators $K^\pm$. Such a splitting cannot be obtained by averaging over the nonunitary conformal module, which need not be semisimple. Compatibility with the full $SO(4,2)$ action, the global zero-mode split, and the BV pairing remains a field-theory obligation. Boundedness on a completed cylinder space is a separate analytic question.

8.4 Raw polynomial instantiation and measured noncompact defects

Computational-supplement material. This subsection records the coordinate instantiation, measured cutoff defects, and basis-level chain dictionaries behind the transfer theorem. It is retained in the archival monolith, but its record-by-record implementation content is assigned to the computational supplement in the planned major-revision split; the extracted residual paper will retain only the resulting propositions and their precise claim boundaries.

The abstract discussion can be instantiated before choosing an analytic completion. In the finite polynomial conformal category, fixed total compact energy gives the exact rational chain

$$G_n \xrightarrow{A_n} M_n \xrightarrow{B_n} E_n \xrightarrow{C_n} I_n, \tag{192}$$

where the four slots contain the Diff×Weyl ghosts, trace-free metric and trace doublet, Bach equation/metric-antifield row, and Noether identity/ghost-antifield row. The Weyl scalar is written as

$$\omega = \sigma + \frac{1}{4}\partial\cdot\xi, \tag{193}$$

so the vector arrow is the trace-free conformal-Killing operator and the trace is an explicit doublet. The raw matrices commute with all four coordinate translations and special-conformal vector fields, not only with $D \times SO(4)$.

The interpretation of these rows can be derived directly from the minimal master action rather than assigned after the calculation. With conventional BV variables $(h_{\mu\nu}, c^\mu, \omega)$ and antifields $(h^{*\mu\nu}, c^*_\mu, \omega^*)$, take

$$S_{\rm BV} = S_W[g] + \int g^{*\mu\nu}\big(\mathcal{L}_c g_{\mu\nu} + 2\omega g_{\mu\nu}\big) + \int c^*_\mu \frac{1}{2}[c,c]^\mu + \int \omega^* \mathcal{L}_c \omega. \tag{194}$$

At the conformally flat background, the tangent differential of its quadratic part has the four-row form

$$(c,\omega) \xrightarrow{K} (h) \xrightarrow{B_{\rm lin}} (h^*) \xrightarrow{K^\sharp} (c^*, \omega^*). \tag{195}$$

Here tangent-complex degree is minus conventional BV ghost number: the four rows have degrees $-1, 0, 1, 2$, whereas the underlying BV coordinates have ghost numbers $1, 0, -1, -2$. In a flat conformal chart, one consistent antibracket convention gives

$$\begin{aligned} Qh_{\mu\nu} &= 2\partial_{(\mu}c_{\nu)} + 2\omega\eta_{\mu\nu}, & Qh^*_{\mu\nu} &= B_{\rm lin}[h]_{\mu\nu}, \\ Qc^*_\nu &= -2\partial^\mu h^*_{\mu\nu}, & Q\omega^* &= 2h^{*\mu}{}_\mu. \end{aligned} \tag{196}$$

The nonlinear terms in (194), including $c^*[c,c]$, $\omega^*\mathcal{L}_c\omega$, and the field-dependent part of $h^*\mathcal{L}_c h$, do not belong to this free tangent differential.

Define the invertible component redefinitions

$$\begin{aligned} &\Omega = \omega + \frac{1}{4}\partial\cdot c, && \tau = \frac{1}{8}\operatorname{tr} h, && h_0 = h - \frac{1}{4}\eta \operatorname{tr} h, \\ &\tau^* = 2\operatorname{tr} h^*, && h_0^* = h^* - \frac{1}{4}\eta\operatorname{tr} h^*, && i^*_\mu = -\frac{1}{2}\left(c^*_\mu + \frac{1}{4}\partial_\mu\omega^*\right), && \Omega^* = \omega^*. \end{aligned} \tag{197}$$

Then (195) becomes exactly

$$(c,\Omega) \longmapsto (K_0 c, \Omega), \qquad (h_0,\tau) \longmapsto (B_{\rm lin}h_0, 0), \qquad (h_0^*, \tau^*) \longmapsto (\partial\cdot h_0^*, \tau^*), \tag{198}$$

which is the raw chain (192). In particular, $\Omega \to \tau$ and $\tau^* \to \Omega^*$ are displayed unit contractible pairs.

Computational Proposition 8.3 (Minimal field-theoretic realization of the raw detour complex)**.** *For each D-finite, $SO(4)$-finite polynomial block in the complete centered buffer $n = 2,3,4,5$, the redefinitions* (197) *give exact rational inverse maps*

$$F_n : C^{\rm min}_{\rm BV,n} \longrightarrow C^{\rm raw}_n, \qquad G_n : C^{\rm raw}_n \longrightarrow C^{\rm min}_{\rm BV,n} \tag{199}$$

satisfying

$$F_n Q_{\rm BV} = Q_{\rm raw}F_n, qquad G_n Q_{\rm raw} = Q_{\rm BV}G_n, qquad F_nG_n = G_nF_n = 1. \tag{200}$$

Thus the comparison homotopies vanish: this phase is a chain isomorphism, not merely a rank-preserving quasi-isomorphism. If $P_{\rm tr}$ selects $(\Omega,\tau,\tau^*,\Omega^*)$ and $s_{\rm tr}$ reverses the two unit arrows, the same matrices verify

$$[Q_{\rm raw}, P_{\rm tr}] = 0, \qquad Q_{\rm raw}s_{\rm tr} + s_{\rm tr}Q_{\rm raw} = P_{\rm tr}. \tag{201}$$

Consequently, in the selected algebraic blocks,

$$\boxed{C^{\text{min,alg}}_{\text{BV,n}} \cong C^{\text{tf,min}}_{\text{raw,n}} \oplus C_{\text{tr},n}, \qquad H(C_{\text{tr},n}) = 0.} \tag{202}$$

The block dimensions are

n	$\dim G_n$	$\dim M_n$	$\dim E_n$	$\dim I_n$	$\dim C_n$
2	90	100	0	0	190
3	160	200	0	0	360
4	259	350	10	1	620
5	392	560	40	8	1000.

(203)

The same triangular trace change is invertible on the inhomogeneous low-mode gauge space and maps the full fifteen-dimensional conformal-Killing kernel to $\Omega = 0$ without changing its rank. More precisely, the exact low-mode matrices give

$$T\big(\ker K_{\text{BV}}\big) = \ker K_{\text{raw}} \cong \mathfrak{so}(4,2). \tag{204}$$

Hence no residual conformal zero mode is created, removed, or absorbed into the trace contraction.

The executable

`symbolic/verify_conformal_field_bv_dictionary.py`

constructs both sides independently, verifies every matrix identity in (200), and emits a 2170-record table with one entry per raw basis vector, including conventional ghost number, antifield number, compact degree, $SO(4)$ content, differential image, and an exact equality flag. This proposition identifies the *minimal quadratic chain*. By itself it does not derive a chosen gauge-fixed nonminimal domain, replace both sides of the global reducibility sector, inventory all additional BV rows, or transfer the field-theoretic cyclic pairing.

The first of these qualifications can be closed without choosing an analytic completion, but it is an implementation corollary rather than a premise of the gauge-invariant classical theorem. Adjoin vector and scalar antighost–multiplier pairs and their antifields,

$$(\bar{c}_\mu, b_\mu; \bar{c}^{*\mu}, b^{*\mu}), \qquad (\bar{\omega}, b_\omega; \bar{\omega}^*, b^*_\omega), \tag{205}$$

with the standard coordinate-BRST rules $s\bar{c} = b$, $sb = 0$ and their Weyl counterparts. The suspended tangent complex used in (195) has the transposed arrows $b \to \bar{c}$ and $\bar{c}^* \to b^*$. Both conventions are retained in the machine dictionary.

In trace-split variables choose the conformal Landau fermion

$$\Psi_{\text{gf}} = \int_M \sqrt{|\bar{g}|}\,\left(\bar{c}^\mu_\perp \bar{\nabla}^\nu h^0_{\mu\nu} + \bar{\omega}\tau\right), \qquad \bar{c}_\perp = P_\perp \bar{c}. \tag{206}$$

It has ghost number -1, is a weight-four scalar density, and commutes with $D \times SO(4)$. No multiplier-square term is used: all multipliers remain in the displayed complex and are removed only by an explicit homotopy. The canonical transformation is

$$T_\Psi : \quad \phi^*_A \longmapsto \phi^*_A + \frac{\delta\Psi_{\text{gf}}}{\delta\phi^A}, \qquad Q_{\text{gf}} = T_\Psi Q_{\text{ext}} T_\Psi^{-1}. \tag{207}$$

For the vector condition the two Euler derivatives are $\delta\Psi/\delta\bar{c} = \bar{\nabla}\cdot h_0$ and $\delta\Psi/\delta h_0 = -K_0\bar{c}/2$ in the conventions of (196); the trace derivatives are the two unit maps $\tau \leftrightarrow \bar{\omega}$.

Computational Proposition 8.4 (Gauge-fixed/nonminimal equivalence)**.** *Let $C_n^{\mathrm{ext}} = C_{\mathrm{BV,n}}^{\mathrm{min}} \oplus C_{\mathrm{nm},n}$, where $C_{\mathrm{nm},n}$ contains the complete Diff and Weyl nonminimal pairs and their antifield duals. In every centered block $n = 2, 3, 4, 5$, the Hessian of (206) gives an exact unipotent map $T_{\Psi,n} = 1 + N_n$, with $N_n^2 = 0$, such that*

$$Q_{\mathrm{gf},n} = T_{\Psi,n} Q_{\mathrm{ext},n} T_{\Psi,n}^{-1}, \qquad Q_{\mathrm{gf},n}^2 = 0. \tag{208}$$

If $p_0, \jmath_0, s_0$ are the direct-sum contraction of the nonminimal pairs, then

$$p_{\mathrm{gf}} = p_0 T_\Psi^{-1}, \qquad \jmath_{\mathrm{gf}} = T_\Psi \jmath_0, \qquad s_{\mathrm{gf}} = T_\Psi s_0 T_\Psi^{-1} \tag{209}$$

obey

$$p_{\mathrm{gf}} \jmath_{\mathrm{gf}} = 1, \qquad \jmath_{\mathrm{gf}} p_{\mathrm{gf}} = 1 - Q_{\mathrm{gf}} s_{\mathrm{gf}} - s_{\mathrm{gf}} Q_{\mathrm{gf}}. \tag{210}$$

Consequently,

$$\boxed{C_{\mathrm{BV,n}}^{\mathrm{gf,alg}} \simeq C_{\mathrm{BV,n}}^{\mathrm{min,alg}} \oplus C_{\mathrm{nm,triv},n}, \qquad H(C_{\mathrm{nm,triv},n}) = 0.} \tag{211}$$

The certified dimensions are

n	$\dim C_n^{\mathrm{min}}$	$\dim C_n^{\mathrm{nm}}$	$\dim C_n^{\mathrm{gf}}$
2	190	52	242
3	360	128	488
4	620	264	884
5	1000	480	1480.

(212)

On the inhomogeneous low-mode block, the exact projector P_Z and its fifty-dimensional complement satisfy

$$P_Z Q_{\mathrm{gf}} = Q_{\mathrm{gf}} P_Z, \qquad Q_{\mathrm{gf}} P_Z = 0, \qquad \operatorname{rank} P_Z = 15, \qquad \ker(K|_{\operatorname{im} P_\perp}) = 0. \tag{213}$$

Thus gauge fixing neither absorbs nor duplicates any conformal-Killing mode. No generalized auxiliary field is silently eliminated.

The executable

```
symbolic/verify_conformal_gauge_fixed_equivalence.py
```

constructs the canonical shear and the transported contraction independently in every block. Its 3094-record ledger classifies every minimal field, nonminimal field, multiplier, and antifield coordinate. Proposition 8.4 does not compute the degree-shifting time-slice transgression (220), establish the post-polarization row ledger, or transfer the field-theoretic pairing; those constructions are supplied below. The bulk dual endpoint itself is identified independently in Proposition 8.5.

8.5 The dual endpoint and the BFV degree shift

There is a short exact argument for the dual bulk zero modes. Write the minimal detour chain in the form

$$G \xrightarrow{A} M \xrightarrow{B} E \xrightarrow{C} I, \qquad C = \pm A^\sharp, \tag{214}$$

where the tangent BV structure gives perfect pairings between G and I and between M and E. Let $Z = \ker A$.

Proposition 8.5 (Canonical dual endpoint)**.** *The endpoint quotient pairs perfectly with the residual gauge kernel:*

$$\boxed{\Theta : I/\operatorname{im} C \xrightarrow{\sim} Z^*, \qquad \Theta([u])(z) = \langle z, u\rangle_{G,I}.} \tag{215}$$

In the exact conformal-Killing block,

$$\dim Z = \dim \operatorname{coker} C = 15, \qquad Z = 4_{-1} \oplus 7_0 \oplus 4_{+1}, \qquad Z^* = 4_{+1} \oplus 7_0 \oplus 4_{-1}. \tag{216}$$

Proof. For $z \in \ker A$, $\langle z, Ce\rangle = \pm\langle Az, e\rangle = 0$, so Θ is well defined. If $\Theta([u]) = 0$, then $u \in (\ker A)^\perp$. Finite-dimensional adjoint linear algebra gives $(\ker A)^\perp = \operatorname{im} A^\sharp = \operatorname{im} C$, proving injectivity; equality of dimensions gives surjectivity. $\square$

The machine certificate does more than compare dimensions. If Z is the 65×15 matrix of the already certified conformal-Killing basis and J_{GI} is the endpoint pairing, it constructs

$$\Theta = Z^\dagger J_{GI}, \qquad J_{GI} C = A^\dagger J_{ME}, \tag{217}$$

verifies $\Theta C = 0$, and produces an exact section S with $\Theta S = I_{15}$. It also verifies

$$I = \operatorname{im} C \oplus S(Z^*) \tag{218}$$

by an exact rank-65 decomposition. These identities and the dual compact grading are generated by `symbolic/verify_conformal_dual_zero_modes.py`.

The interpretation requires one further distinction. The bulk endpoint class, the residual BFV momentum, and the constraint value all transform in dual modules, but they are not the same coordinate:

object	space	degree	role
c^a	$Z[1]$	$+1$	residual ghost
b_a	$Z^*[-1]$	-1	BFV ghost momentum
μ_a	Z^*-valued functions	0	constraint
$[u]$	$\operatorname{coker} C \cong Z^*$	bulk endpoint	obstruction codomain

(219)

The time-slice BV–BFV construction must supply a degree-shifting transgression

$$\tau : H^{\text{bulk}}_{\text{endpoint}} \longrightarrow Z^*[-1]_{\text{BFV}}, \tag{220}$$

in the sense of the bulk-to-boundary BV–BFV correspondence [24]. Equivariance reduces it to one scalar, $\tau = \lambda\Theta$, after the generator bases are fixed. In the selected algebraic BFV convention this scalar is no longer free. Normalize endpoint representatives by $\Theta([u_a]) = e^a$. The endpoint/Taub theorem and the canonical BFV charge give, on the ghost-free slice,

$$Q_{\text{bulk}} u_D = \mu_D, \qquad Q_{\text{BFV}} b_D = \mu_D, \qquad b_D = \lambda u_D, \tag{221}$$

so compatibility forces $\lambda = +1$ for $\Omega_{\text{gh}} = \delta b_a \wedge \delta c^a$ and the sign of Ω_{res} displayed below. Equivariance extends the result to all fifteen components. Reversing either canonical sign reverses λ but cannot leave an arbitrary magnitude. The exact role audit verifies one residual ghost copy, one endpoint quotient, one $15 + 15$ BFV cotangent sector, and refuses to identify $[u]$ with b_a without the degree suspension (220); see `symbolic/verify_conformal_residual_bfv_roles.py`. The normalized suspension, homogeneous generator order, cotangent orientation, and centered overlap are certified independently by `symbolic/verify_conformal_zero_mode_transgression.py`.

Schematically, the no-duplication map is a transfer, not a deletion:

$$\begin{array}{ccc} Z_{\mathrm{local}} \oplus H^{\mathrm{bulk}}_{\mathrm{endpoint}} & \subset & C^{\mathrm{min}}_{\mathrm{BV}} \\ \big\downarrow{\scriptstyle \mathrm{id}\oplus\tau} & & \big\downarrow \\ Z[1] \oplus Z^{*}[-1] & \subset & C^{\mathrm{res}}_{\mathrm{BFV}}. \end{array} \tag{222}$$

The moment map is a function into Z^* and is not a third set of zero-mode coordinates.

8.6 The free symmetric-algebra state complex is obtained after polarization

The full cyclic bulk BV tangent complex should not be asserted to have only one cohomological row: its ghost, physical, and antifield sectors occur with their cyclic duals. The single-row criterion used by the residual calculation belongs after the sequence

$$C^{\mathrm{bulk}}_{\mathrm{BV}} \longrightarrow C^{\Sigma}_{\mathrm{BFV}} \longrightarrow C^{+}_{\mathrm{state}}, \tag{223}$$

where the first arrow is time-slice BV–BFV transgression and the second is canonical phase-space reduction followed by the positive-frequency Lagrangian polarization. Nonminimal quartets contract in this passage; negative-frequency modes supply symplectic duals rather than additional positive-energy coefficient states. The final coefficient complex is the polarized free symmetric-algebra (Fock) realization

$$\mathcal{F}(\mathcal{W}_{+} \oplus \mathcal{W}_{-}) \otimes \Lambda^{\bullet}\mathfrak{g}^{*}, \tag{224}$$

not the unpolarized bulk BV antifield complex. Here "Fock" names the bosonic symmetric-algebra construction associated with the chosen positive-frequency Lagrangian subspace; it does not assert a positive one-particle Hilbert metric.

Correspondingly, the even form on the positive-energy module is induced from the real phase-space symplectic form,

$$\Omega_{\mathrm{BV}} \longrightarrow \Omega_{\mathrm{BFV}} \longrightarrow \Omega_{\Sigma} \longrightarrow J_{+}(u,v) = -i\Omega_{\Sigma}(\bar{u},v), \tag{225}$$

in the convention (119); reversing the convention reverses both displayed signs. This is distinct from restricting the odd BV form to one positive-frequency half. A complete row ledger must therefore be generated after the two arrows in (223). The selected algebraic ledger and pairing are constructed below.

Computational Proposition 8.6 (Raw metric-BV retract and conformal transfer)**.** *In the complete energy buffer $n = 2,3,4,5$, exact rational changes of basis give maps $p_n, \jmath_n, s_n$ satisfying*

$$p_n \jmath_n = 1, \qquad \jmath_n p_n = 1 - q_n s_n - s_n q_n, \tag{226}$$

with cohomology dimensions $10, 40, 82, 136$. The chosen retract is not strictly equivariant under the noncompact generators. For one Cartesian component, the defect ranks are

	rank ρ_H	rank $\delta_\jmath$	rank δ_p	rank δ_s
P_0	40	29	5	35
K_0	82	80	9	97.

(227)

Every inclusion defect is explicitly q-exact. Each projection defect has an explicit right q-homotopy, while the homotopy defect obeys the induced projector identity. The transferred maps obey all sixteen $[K_b, P_a]$ brackets exactly on cohomology. Moreover,

$$p\rho_a s\rho_b \jmath = 0 \tag{228}$$

on the physical metric row for every pair of proper-conformal components; all higher terms vanish because a second s acts on the gauge row.

Thus full equivariance of an arbitrarily chosen representative inclusion is not needed and, for this natural exact splitting, is false. What transfers strictly is the conformal action on cohomology. The calculation therefore realizes the homotopy-equivariant outcome anticipated above rather than silently replacing it by a strict SDR. The exact matrices, defect homotopies, and generated table are produced by `symbolic/verify_conformal_raw_bv_transfer.py`.

Computational Proposition 8.7 (Raw cross-energy cyclic form)**.** *Let J_2 be the positive energy-two form pulled back from the electric and magnetic Weyl curvature. In the raw polynomial cohomology basis the four raising blocks have joint full row rank, and the recursion*

$$J_n K^+_{a,n-1} = -(K^-_{a,n})^\dagger J_{n-1}, \qquad a = 1, \ldots, 4, \tag{229}$$

determines a unique nondegenerate J_n at every subsequent energy. Through the complete centered buffer its exact inertias are

$$\begin{array}{c|cccc} n & 2 & 3 & 4 & 5 \\ \hline (n_+, n_-) & (10, 0) & (24, 16) & (42, 40) & (64, 72). \end{array} \tag{230}$$

These are precisely the parity-complete $+E, -A, -L$ signatures.

Transporting the form through the exact raw adapted basis makes $q, s, \jmath, p$ cyclic in the ghost, metric, equation-antifield and identity-antifield rows. Every elementary representative dressing $s\rho_A\jmath$ lies in their common isotropic gauge subspace. Consequently all mutual quadratic dressing terms vanish and

$$I^\sharp I = 1 \tag{231}$$

for the transferred inclusion in the centered window.

The form is obtained from (229), not from a postulated $E/A/L$ change of basis. Exact rational congruence reduction computes the displayed inertias without numerical eigenvalues. The certificate is `symbolic/verify_conformal_cross_energy_pairing.py`.

Computational Proposition 8.8 (Complete centered HPL correction test)**.** *For every allowed ordered pair of the fifteen residual generators on source energies two, three, and four which remains in the coefficient buffer,*

$$p\Delta s\Delta\jmath = 0, \qquad s\Delta s\Delta\jmath = 0. \tag{232}$$

The second identity holds before projection and kills every higher physical row term. The universal ghost-structure term acts trivially on the local rows and $s\jmath = 0$. The only path beyond the buffer is double raising from energy four; it vanishes by the four-raising-ghost exterior saturation of the centered top shell. Hence

$$\boxed{Q_H = p\Delta\jmath = d_{\mathrm{CE}}} \tag{233}$$

in the complete centered coefficient window.

This check contains 555 compact, lowering, raising, and mixed ordered pairs; it is stronger than the proper-conformal pair test in (228). Its machine report is generated by `symbolic/verify_conformal_full_hpl_transfer.py`.

8.7 The filtered local-to-residual complex

The possible failure modes can be displayed without compressing them into the phrase "no other row." This filtration is formed *after* the time-slice transfer, canonical phase-space reduction, positive-frequency polarization, and nonminimal-quartet contraction in (223); it is not the cohomology spectral sequence of the full unpolarized cyclic bulk tangent complex. After the conformal-Killing zero modes have been transferred from the local ghosts, introduce the residual-ghost filtration on the polarized state-resolution bicomplex

$$C^{p,q} = \Lambda^p \mathfrak{g}^* \widehat{\otimes} C^q_{\text{local}}, \qquad F^p C = \bigoplus_{r \geq p} C^{r,*}. \tag{234}$$

Here p is residual ghost number and q is the degree in the remaining polarized local resolution. Antifields and negative-frequency conjugates which supplied the bulk cyclic duals are not silently deleted: their role must already have been accounted for by the two arrows in (223). A transferred degree-one differential has the filtered expansion

$$Q_H = q_{\text{local}} + d_{\text{res}} + \sum_{k \geq 2} Q_{[k]}, \tag{235}$$

with bidegrees

$$|q_{\text{local}}| = (0,1), \qquad |d_{\text{res}}| = (1,0), \qquad |Q_{[k]}| = (k, 1-k). \tag{236}$$

The terms $Q_{[k]}$ encode higher transferred ghost operations; assuming that they vanish is a substantive strictness assertion, not automatic homological bookkeeping.

The resulting spectral sequence begins

$$E_1^{p,q} = \Lambda^p \mathfrak{g}^* \otimes H^q(q_{\text{local}}), \qquad\qquad E_2^{p,q} = H^p(\mathfrak{g}; H^q(q_{\text{local}})), \tag{237}$$

and

$$d_r : E_r^{p,q} \longrightarrow E_r^{p+r, q-r+1}. \tag{238}$$

For the desired term $E_r^{4,0}$, a higher differential can enter from $(4-r, r-1)$ for $r = 2, 3, 4$, and can leave toward $(4+r, 1-r)$ for $2 \leq r \leq 11$. Thus local classes in degrees $q = 1, 2, 3$ and antifield or negative-degree classes down to $q = -10$ are potential modifiers. They must be inventoried or excluded; it is not enough to compute only the local degree-zero row. Since $0 \leq p \leq 15$, this filtration is bounded. On each finite compact-energy block the cochain spaces used here are finite, so the spectral sequence converges. An infinite completed state space additionally requires a complete, separated filtration. Even after collapse, a cyclic splitting is needed to control extension data and the Hermitian form.

This is the standard local-BRST setting in which antifield and gauge-fixed cohomologies must be compared carefully [19, 21].

Proposition 8.9 (Single-row strictness and collapse)**.** *For the polarized state-resolution bicomplex just described, suppose the global zero modes have been transferred exactly,* $H^q(q_{\text{local}}) = 0$ *for* $q \neq 0$*, and the free residual perturbation* Δ *has bidegree* $(1,0)$*. Suppose in addition that the chosen SDR is compatible with the local grading, so that* s *has bidegree* $(0,-1)$ *and* $\jmath, p$ *are supported in local degree zero. Then*

$$\boxed{Q_H = p \Delta \jmath,} \tag{239}$$

all higher HPL terms vanish, and the local-to-residual spectral sequence collapses to

$$H^p\big(\mathfrak{g}; H^0(q_{\text{local}})\big). \tag{240}$$

Proof. The homotopy s lowers local degree by one. Since both $\jmath(H)$ and the image retained by p lie in local degree zero,

$$p\Delta(s\Delta)^r\jmath = 0 \qquad (r \geq 1). \tag{241}$$

The HPL series therefore reduces to (239). The E_1 page has only the row $q = 0$, so no higher differential can enter or leave it. □

The proposition concerns the perturbation obtained from the *free metric BV bicomplex*. It must not be inferred by naively restricting a nonlinear minimal Q-manifold. In the Dneprov–Grigoriev variables, curvature-dependent terms schematically of the form $\xi\xi W$ and $\xi\xi C$ occur in ghost transformations and can have total compact degree zero [27]. They belong to the nonlinear comparison and must either be identified with the interacting deformation or removed by the equivalence back to the free metric presentation before Proposition 8.9 is applied.

Proposition 8.10 (Formal filtered-transfer criterion)**.** *Suppose the zero-mode split produces* (234)*; the filtration is convergent;* $H^q(q_{\mathrm{local}})$ *vanishes in every row that can enter or leave* $(4,0)$*; and the transferred differential on* $H^0(q_{\mathrm{local}})$ *is the strict residual Chevalley–Eilenberg differential, with all relevant* $Q_{[k]}$ *absent. Then the full cohomology in the selected filtered degree is isomorphic, as a vector space, to the residual group*

$$H^4\big(\mathfrak{g}; H^0(q_{\mathrm{local}})\big)\,. \tag{242}$$

If in addition the retract is compact-degree equivariant and cyclic, the Cartan homotopy and the cohomological Hermitian pairing transfer, and the dressed inclusion is the isometry (191)*.*

Proof. Under the stated row and strictness assumptions no differential (238) enters or leaves $(4,0)$, so the bounded spectral sequence collapses there to its E_2 value. The equivariant and cyclic statements are (188) and (191). The cyclic splitting fixes the extension of the pairing from the associated graded object. □

8.8 End-to-end centered integration in the polynomial category

The transferred coefficient matrices of Proposition 8.6 can be coupled directly to the independently generated residual BFV/CE ghosts. This removes the hand-specified $E/A/L$ matrices from the final centered rank calculation.

Computational Proposition 8.11 (Metric-to-residual integration)**.** *In the parity-complete algebraic polynomial complex, the pipeline*

$$\textit{metric BV rows} \longrightarrow H(q) \longrightarrow C^\bullet(\mathfrak{so}(4,2); H(q)) \tag{243}$$

has the exact centered dimensions and ranks

	$\dim C^3$	$\dim C^4$	$\dim C^5$	$\operatorname{rank} d_3$	$\operatorname{rank} d_4$	$\dim H^4$
vacuum	147	407	833	116	291	0
one particle	580	2622	7314	520	2102	0
two particles	0	55	385	0	53	2.

(244)

For the two-particle row, the next space has dimension 1155 *and* $d_5d_4 = 0$ *exactly. Orientation reversal acts on the two kernel vectors by*

$$\mathrm{diag}(-1, +1). \tag{245}$$

Pulling the positive energy-two electric/magnetic curvature form back to the metric representatives gives the raw kernel Gram matrix

$$\operatorname{diag}\left(\frac{5}{64}, 5\right). \tag{246}$$

After exact normalization and multiplication by the intrinsic unit residual ghost norm, the representative Gram matrix is I_2.

The odd and even eigenvectors are respectively the Pontryagin and Weyl-square directions. The calculation verifies the vector-space transfer and a positive-definite representative form on this two-dimensional cohomology, normalized to I_2, in the algebraic model. Proposition 8.7 independently fixes its cross-energy cyclic normalization and dressed isometry. The certificate and machine-readable ranks are produced by

`symbolic/verify_conformal_metric_to_residual_integration.py`.

8.9 The selected closed-universe BFV problem

The residual quotient is now specified as part of the theorem's input. The Cauchy surface is the closed oriented S^3, no boundary-charge variables or external clock are adjoined, and all fifteen conformal reducibilities are represented by BFV constraints. In particular,

$$\partial S^3 = \varnothing, \qquad \operatorname{rank} Q_\partial = 0, \qquad D \in \{G_A\}_{A=1}^{15}. \tag{247}$$

The complementary-degree CE pairing then gives Lemma 6.1. This makes the centered polarization and the use of ι_D internally consistent with the chosen boundary problem. These are precisely the closed-universe assumptions entering the derived zero-locus and stabilizer quotient of Theorem 8.13.

Equation (247) is a selection, not a universal theorem. Adding a boundary, clock, asymptotic charge, or deparametrized Hamiltonian removes D from the constraint list and disables the Cartan contraction. The two alternatives are represented explicitly by `symbolic/verify_conformal_closed_universe_bfv.py`.

8.10 The selected algebraic BV–BFV completion

The smooth BGG theorem and Theorem 2.5 remove the global PDE-surjectivity and finite-mode preimage gaps. Propositions 8.6 and 8.11, together with Propositions 8.7 and 8.8, additionally close the kernel, image, noncompact action, cyclic form, transferred differential, and centered-rank calculation in an explicit polynomial metric-BV model. Proposition 8.3 also identifies its four minimal rows exactly with the tangent complex derived from the quadratic master action. Proposition 8.4 further derives a concrete Landau-gauge canonical transformation, retains both nonminimal antifield duals, contracts the entire nonminimal extension, and proves that the fifteen conformal-Killing modes are unchanged. Proposition 8.5 now also identifies the bulk endpoint quotient canonically with Z^*, while (133) identifies the quadratic endpoint obstruction with the certified Taub/moment map. In the selected algebraic BV–BFV polarization these ingredients now close.

Theorem 8.12 (Selected algebraic BV–BFV polarization)**.** *Work on the D-finite, $SO(4)$-finite cylinder. Choose the closed-universe BFV problem (247) and the positive-frequency Lagrangian polarization. Then:*

1. *the normalized endpoint suspension is*

$$\boxed{\tau = \Theta, \qquad \lambda_D = \lambda_{\text{all}} = +1,} \tag{248}$$

and transfers, rather than copies, the local zero-mode cotangent pair into one residual pair (c^a, b_a)*;*

2. *the polarized free symmetric-algebra state complex is exactly*

$$\boxed{\mathcal{H}_{\text{state}} = \mathrm{Sym}(\mathcal{W}_+ \oplus \mathcal{W}_-) \otimes \Lambda^\bullet \mathfrak{so}(4,2)^*,} \tag{249}$$

with local nonzero-mode cohomology concentrated in polarized degree zero;

3. *the induced Hermitian field pairing on the two centered residual classes is positive-definite and, in the displayed normalized basis, is*

$$\boxed{\langle W_r^2 v_-, W_s^2 v_- \rangle_{\text{field}} = \delta_{rs}, \qquad G_{\text{field}} = I_2.} \tag{250}$$

No single-row or Fock-space assertion is made about the unpolarized classical bulk BV tangent complex.

Proof. Choose endpoint representatives u_a by $\Theta([u_a]) = e^a$. Equation (133) gives $Q_{\text{bulk}} u_a = \mu_a$ in the quotient. For $\Omega_{\text{gh}} = \delta b_a \wedge \delta c^a$ and

$$\Omega_{\text{res}} = c^a \mu_a - \frac{1}{2} f^a{}_{bc} c^b c^c b_a, \tag{251}$$

the ghost-free Hamiltonian vector field gives $Q_{\text{BFV}} b_a = \mu_a$. The D component of $b_a = \lambda u_a$ therefore forces $\lambda = 1$; coadjoint equivariance gives all fifteen components. In the ordered $4_{-1} \oplus 7_0 \oplus 4_{+1}$ basis, the induced determinant orientation agrees with $v_+ \wedge \Theta_0 \wedge v_-$ and fixes $(v_-, v_-)_{\text{gh}} = 1$.

The complexified physical phase space splits as $L_+ \oplus L_-$, where $L_+ = \mathcal{W}_+ \oplus \mathcal{W}_-$ and L_- is its symplectic conjugate. The exact cross-energy form makes both halves Lagrangian and induces $J_+(u, v) = -i\Omega_\Sigma(\bar{u}, v)$. The all-energy conformal action preserves L_+. Every local, trace, and nonminimal complement is an explicit differential doublet. On its polynomial state algebra the standard number-operator contraction leaves only the vacuum. Finally c^a acts by exterior multiplication and b_a by ι_a, so no second ghost exterior algebra or endpoint state row is present. This proves (249).

The action-normalized energy-two E primary has unit symplectic norm. An exact energy-two isometry identifies this form with the raw curvature pullback J_2; the contravariant recursion (229) fixes every higher energy. Multiplying the resulting normalized two-class matter Gram matrix by the unit oriented ghost overlap yields (250). The exact constructions and sector ledger are generated by the three executables

`symbolic/verify_conformal_zero_mode_transgression.py`,
`symbolic/verify_conformal_polarized_state_complex.py`,
`symbolic/verify_conformal_polarized_pairing_transfer.py`.

□

Theorem 8.13 (Derived residual reduction for the selected closed universe)**.** *Let $\mathcal{P}_{\mathrm{lin}}$ denote the formal linearized pure-Weyl phase space after quotienting local Diff$\times$ Weyl transformations with vanishing Cauchy data. Let*

$$\mathfrak{g} = \mathrm{Lie}\,\mathrm{Stab}_{\mathrm{gauge}}(\bar{g}) \cong \mathfrak{so}(4,2) \tag{252}$$

be the infinitesimal residual stabilizer of the cylinder representative, and let $\mu : \mathcal{P}_{\mathrm{lin}} \to \mathfrak{g}^*$ be the quadratic moment map normalized by (127). Assume that

1. *$\Sigma = S^3$ is closed and carries no boundary or asymptotic charge;*
2. *no external clock is adjoined and the cylinder is not deparametrized;*
3. *all fifteen residual reducibilities, including D, are gauged;*
4. *one works formally near $\bar{g}$ in the zero-charge, second-order integrable sector; and*
5. *the residual action is treated infinitesimally, without assuming a free group action or an integrated $SO(4,2)$ representation.*

Then, in the selected formal algebraic category constructed in this paper, the residual classical problem is represented by the derived Hamiltonian reduction

$$\boxed{\mathcal{P}_{\mathrm{der}} = \left[\mathcal{P}_{\mathrm{lin}} \overset{\mathbf{R}}{\underset{\mathfrak{g}^*}{\times}} \{0\} /\!/^{\mathbf{R}} \mathfrak{g}\right].} \tag{253}$$

Equivalently, it is represented by the residual BFV complex with coordinates c^a and b_a of degrees $+1$ and -1, whose differential satisfies

$$Qf = c^a\{\mu_a, f\}, \qquad Qc^a = -\frac{1}{2} f^a{}_{bc} c^b c^c, \qquad Qb_a\big|_{c=0} = \mu_a. \tag{254}$$

After the BV–BFV time-slice map and the selected positive-frequency polarization, this becomes the absolute complex (137) on the free symmetric-algebra coefficient module.

Proof. Proposition 4.1 proves that every second-order integrable perturbation lies in the zero locus of the fifteen quadratic Taub charges. Equation (127), with the all-energy normalization of Proposition 4.2, identifies those charges componentwise with the equivariant Hamiltonian moment map μ. Thus the equation imposed by each residual BFV momentum is the actual linearization-stability constraint, rather than a new algebraic restriction.

The residual kernel is $Z = \ker K = \ker A \cong \mathfrak{g}$ in the notation of the minimal detour chain. Proposition 8.5 and (215) give the perfect pairing $\operatorname{coker} K^\sharp \cong Z^* \cong \mathfrak{g}^*$; the role separation in (219) distinguishes this endpoint quotient, the constraint value μ, and the BFV momentum. The time-slice map (220) then supplies the required degree suspension. The normalized identity (248), proved in Theorem 8.12, shows that it produces exactly one cotangent pair (c^a, b_a) with degrees $(+1, -1)$ and no duplicate endpoint sector.

With the canonical ghost symplectic form, the residual BFV charge (251) generates (254). The restriction $c = 0$ in its last equation isolates the Koszul term; the full Qb_a also contains the coadjoint ghost term. Equivariance of μ and the Jacobi identity give $Q^2 = 0$. Consequently the Koszul part is the formal derived zero fibre of μ, while the ghost part is the Chevalley–Eilenberg derived quotient by the infinitesimal stabilizer. This is precisely (253); nonregular stabilizer directions are retained rather than replaced by an ordinary manifold quotient [24].

Finally, item 2 of Theorem 8.12 identifies the polarized coefficient algebra with (249). There c^a acts by exterior multiplication and b_a by ι_a, so the BFV vector field generated by (251) is exactly the absolute differential (137). This proves the last assertion. The argument is formal and infinitesimal in the selected algebraic category; it neither constructs a global singular derived stack nor identifies the unpolarized classical BV tangent complex with a graviton Fock space. □

Remark 8.14 (Fixed cylinder and relative alternatives). In the fixed-background cylinder problem the moment-map values are global charges rather than constraints: one neither restricts to $\mu^{-1}(0)$ nor quotients by the full stabilizer, and the coefficient module remains an $SO(4,2)$ module with D as Hamiltonian. The relative primary kernel (149) is useful representation-theoretic data, but is not the derived reduction (253). Gauging only a subgroup, retaining boundary charges, or deparametrizing time requires the corresponding relative or boundary BFV complex; it is not obtained by deleting the D ghost from the absolute complex.

Corollary 8.15 (Algebraic strict-pure-Weyl result)**.** *For the selected algebraic closed-cylinder BV–BFV data, the complete free/linearized polarized residual state cohomology in the selected degree is* (167)*. Its field-induced Hermitian form is positive-definite and normalized to I_2, and its local Euler–Lagrange quotient is* (181) *with form I_1.*

Proof. Theorems 2.4 and 2.5 identify the one-particle module, while Proposition 8.1 gives its free Fock lift. The minimal field/raw dictionary and gauge-fixed contraction exhaust the local algebraic variables. Theorem 8.12 supplies the one-copy residual BFV sector and its pairing. Propositions 8.6, 8.8, and 8.11 then give the strict residual differential, the centered kernel and image, and the two normalized classes. □

This theorem is deliberately category-limited. It constructs the selected finite algebraic degree suspension characterized by endpoint duality and compatibility of the two cohomological vector fields; it is not an independent continuity theorem for a temporal boundary one-form. It also does not prove closed range or uniqueness among other boundary polarizations. A Hilbert, Krein, distributional, or completed Fock statement requires separate analysis.

9 Scope, interpretation, and open questions

Residual gauge choice. The result is specific to quotienting the full residual $SO(4,2)$ action on the compact cylinder. If some generators are retained as global or asymptotic charges, the physical complex is different. In particular, Cartan contraction cannot be used to erase nonzero D energy when D is the physical Hamiltonian.

No positive graviton Fock claim. The unreduced module remains indefinite and the one-particle residual cohomology is zero. The positive-definite pairing belongs to two weight-four residual vertex/deformation classes. Calling I_2 a positive graviton Hilbert space would therefore misidentify both the cohomological degree and the state–operator map.

Classical BV data and the free state representation. The local BV–BFV differential, current, and residual moment map are classical linearized data. Choosing a positive-frequency Lagrangian polarization and completing the symmetric algebra produces the selected free BRST state representation; it is already a free quantization choice, not an unpolarized classical phase space. The result does not show that interactions preserve the reduced space or pairing, and it does not establish the quantum master equation. A local Diff×Weyl anomaly can obstruct quantum nilpotency even though Corollary 8.15 and Theorem 2.6 hold in the stated free/linearized category.

Separate quantum local-algebra status. A separate `LOCAL-ALGEBRAIC` programme classifies curvature counterterm and anomaly candidates. It currently supplies invariant-basis reductions at derivative orders four and six, Weyl/Schouten and Hodge-sign checks, the universal diffeomorphism descent, the complete intrinsic Euler tower, and fail-closed interfaces for importing antifields and forming relative cohomology. Its antifield-zero lower-form grading enumeration now has 720 refined signatures and a complete factored tensor-slot realization; the Bianchi, integration-by-parts, dimension-specific, and sparse-matrix quotients remain open. These results do not enter the Lorentzian causal theorem proved here. In particular, they do not yet establish the production $H^{0,4}(s \mid d)$ or $H^{1,4}(s \mid d)$ quotients, anomaly coefficients, antifield completion, restoration of the quantum master equation, or a Lorentzian quantum theory. Detailed ranks, lifecycle states, and certificate paths are maintained in the companion computational supplement, `quantum-weyl/README.md`, and `quantum-weyl/reports/`, rather than repeated in this paper.

Vertex versus dynamics. The two residual classes do not define two independent local gravitational dynamics. The parity-even class is the Weyl action direction; the parity-odd class is a theta direction, locally related by a canonical boundary transformation. The positive dynamical quotient is therefore one-dimensional.

Bridge, completion, and covariant boundary. The all-level smooth metric-to-$\mathcal{W}_{\mathrm{geom}}$ identification and the algebraic $E/A/L$ curvature intertwiner are exact. The raw polynomial metric-BV complex now supplies its explicit ghost and antifield detour rows, explicit noncompact homotopies, strict induced conformal action, cross-energy cyclic form, vanishing centered HPL corrections, canonical residual CE pairing, and end-to-end centered ranks. Its minimal four-row differential has also been derived from the quadratic field-theoretic master action and identified by exact inverse chain maps. A concrete conformal-Landau gauge fermion and the full vector and scalar nonminimal sectors have now also been canonically adjoined and explicitly contracted, with all fifteen conformal-Killing modes preserved. The selected algebraic endpoint-to-BFV suspension now has $\lambda = +1$; the post-BFV positive-frequency ledger contracts every nonphysical doublet to its vacuum and leaves (249); and the pairing transfer through $\Omega_{\mathrm{BV}} \to \Omega_{\mathrm{BFV}} \to \Omega_{\Sigma} \to J_+$ gives $G_{\mathrm{field}} = I_2$. The direct Taub comparison does not require first identifying a degree-three BGG representative, although that geometric comparison remains valuable. The normalized energy modes now have the Krein–Fock completion of Theorem 2.6, with closed residual operator, closed range, and unchanged H^4. The reduced Lorentzian metric fields have the Cauchy–Sobolev realization of Theorem 2.7, while Proposition 2.8 supplies the exact curved operator identities, support-local BV deformation retract, and covariant current comparison. The null-symbol theorem (60) rules out a scalar-wave witness on the original 24-field bundle and identifies the uncancelled quotient with the two physical helicities; the reduced Weyl symbol (62) is an isomorphism. The exact 26-state Weyl–Cotton equations, their constraint-adjusted symmetric-hyperbolic closure, sourced subsidiary identity and all-level $E/A/L$ spectrum are now proved. The all-row local prolongation, BV differential, and off-shell current comparison are exact. The transferred all-row causal homotopy (102) is exact. Its causal difference is a compact-to-global quasi-isomorphism on the cylinder; cutoff representatives recover all fifteen residual endpoints and their duals without duplication, and the cutoff commutator transfers the $SO(4,2)$ action up to chain homotopy. The direct cyclic Green/current comparison completes the pairing transport. Thus Theorem 2.9 proves the final covariant H^4 statement by transport through (72), not by another residual CE calculation. The two false flags in (103) remain scoped to the canonical endpoint Green inverse and prolonged-witness implementations and are not dependencies of that theorem. Hadamard completion and uniqueness among boundary or deparametrized polarizations remain separate analytic questions. The exact ranks in Table 1 remain independent convention and regression checks.

10 Conclusion

Residual conformal reduction changes the state-space question rather than re-signing the indefinite oscillators of free Weyl gravity. For the selected closed-cylinder polarization, Cartan contraction reduces the absolute residual complex to three matter sectors. The vacuum and one-particle sectors are acyclic, while the two-particle sector gives

$$H^4_{\mathrm{res,min}} = \mathrm{span}\{[W_+^2],[W_-^2]\}, \qquad \operatorname{sig} G_{\mathrm{res}} = (2,0), \qquad G_{\mathrm{res}} = I_2.$$

They are Weyl-square and Pontryagin deformation directions, not one-particle gravitons. The residual Hermitian form is positive-definite and is normalized to I_2 only after the selected basis, orientation, and unit-overlap choices; the variational quotient leaves one dynamical class.

The covariant theorem follows by transport, not by a second residual calculation. The auxiliary and curvature mapping-cylinder retracts are support local, the Weyl–Cotton system carries the complete $E/A/L$ module, and the curved adjoint-tractor plus hybrid contraction gives causal homotopies on all prolonged rows. Their causal difference is a compact-to-global quasi-isomorphism, recovers all fifteen residual endpoints without duplication, transfers $SO(4,2)$ up to the cutoff chain homotopy, and identifies the Green and current pairings. Consequently Theorem 2.9 gives $H^4_{\mathrm{cov}} = \mathrm{span}\{[W_+^2],[W_-^2]\}$ with signature $(2,0)$, normalized to $G_{\mathrm{cov}} = I_2$ after the same choices.

The theorem remains classical, free, and polarization dependent. The rank-11 versus rank-9 null-symbol result rules out only the stated 24-field scalar-wave witness, and the invariant two-wave factorization is excluded only in its certified family. A canonical endpoint Green inverse, a prolonged degreewise Green witness, alternative boundary polarizations, distributional/Hadamard completion, interaction descent, quantum-master restoration, and quantum unitarity are not claimed.

A Conventions

The cylinder radius is one and the Lorentzian signature is $(-,+,+,+)$. Compact energy is the eigenvalue of $D = i\partial_t$ in the oscillator convention. The compact algebra is $SO(4) \cong SU(2)_L \times SU(2)_R$. A label (j_L, j_R) has dimension $(2j_L+1)(2j_R+1)$. Positive chirality means the lowest Weyl primary $(2;2,0)$; parity exchanges j_L and j_R.

The normalized module convention is (115). The reduced curvature action (8) differs by an overall factor and sign from the common $-\int C^2$ oscillator convention. All moment-map comparisons derive the symplectic form and Taub kernel from the same action, so the overall action factor cancels. Lorentzian self-duality may insert a factor of i in the odd parity combination; no conclusion depends on that convention.

The residual generator degrees are $(0^7, -1^4, +1^4)$ and dual ghost degrees are their negatives. Ghost number and compact degree are independent. The selected absolute residual degree in this paper is ghost number four, fixed by (139); it is not ordinary ghost number zero. We call it a candidate physical degree only under the full residual gauging specified in Section 5.2.

B Exact rank certification

This implementation-facing appendix is assigned to the computational supplement in the planned paper split. It remains here until the extracted articles and supplement build and verify independently.

The one-particle matrices contain radicals. They are represented over

$$K = \mathbb{Q}(i, \sqrt{2}, \sqrt{3}, \sqrt{5}). \tag{255}$$

For a lower bound on rank, the calculation maps this field to $\mathbb{F}_{241}$ by

$$i \mapsto 64, \qquad \sqrt{2} \mapsto 22, \qquad \sqrt{3} \mapsto 56, \qquad \sqrt{5} \mapsto 103. \tag{256}$$

Each image obeys its defining square relation. A nonzero minor after this good-prime reduction is a nonzero characteristic-zero minor. Exact nilpotency provides the complementary upper bound

$$\operatorname{rank} d_{q-1} + \operatorname{rank} d_q \leq \dim C^q. \tag{257}$$

For the one-particle complex the modular lower bounds saturate this inequality. For the two-particle complex the two explicit independent kernel vectors provide the upper bound which the modular rank saturates. Thus the ranks quoted in Section 6 are exact characteristic-zero ranks, not probabilistic modular estimates.

C Endpoint factorization no-go receipt

This generated algebraic receipt is supplement material. It remains an active, tracked input of the archival monolith during the split.

For auditability, this appendix records the nonzero part of the exact Nullstellensatz certificate used in (97). The variables $a_0, \ldots, a_8$ are the nine coefficients of the invariant first-order term A_- after the cubic equation has imposed $A_+ = -A_-$. The f_i are the indexed residual quadratic coefficient equations after the three zeroth-order sums have been eliminated; all unlisted multipliers vanish. The table is generated mechanically from `curved_endpoint_metric_factorization_no_go.json`, and its generator re-expands the displayed identity over $\mathbb{Q}$ before writing the TeX file.

$$\begin{aligned} f_0 = &-2a_0a_4 - a_0a_5 \\ &- 16a_1a_4 - 8a_1a_5 \\ &- 8a_2a_4 - 4a_2a_5 \\ &+ 2 \end{aligned}$$

$$\begin{aligned} w_0 = \frac{3}{128}\Big(&18a_0^2 - 192a_0a_1 \\ &- 72a_0a_2 + 23a_0a_6 \\ &+ 384a_1^2 + 192a_1a_2 \\ &+ 62a_1a_6 + 58a_2a_6 \\ &+ a_3a_7 - 152a_4a_6 \\ &+ 224a_4a_8 - 68a_5a_6 \\ &+ 112a_5a_8\Big) \end{aligned}$$

$$
\begin{aligned}
f_1 = {} & 2a_0a_6 - 3a_0a_8 \\
& + 16a_1a_6 - 24a_1a_8 \\
& + 8a_2a_6 - 12a_2a_8 \\
& - 2
\end{aligned}
$$

$$
\begin{aligned}
w_1 = {} & \frac{3}{128}\Big(27a_0a_4 + 24a_0a_5 \\
& - 48a_1a_4 - 60a_1a_5 \\
& + 2a_3^2 + 2a_3a_7 \\
& - 224a_4^2 - 224a_4a_5 \\
& - 56a_5^2 + 6a_7^2 \\
& - 58\Big)
\end{aligned}
$$

$$
\begin{aligned}
f_2 = {} & -a_0a_8 + 6a_1a_6 \\
& - 8a_1a_8 + 2a_2a_6 \\
& - 4a_2a_8 - 2
\end{aligned}
$$

$$
\begin{aligned}
w_2 = {} & -\frac{3}{128}\Big(81a_0a_4 + 72a_0a_5 \\
& - 144a_1a_4 - 180a_1a_5 \\
& + 6a_3^2 + 6a_3a_7 \\
& - 224a_4^2 - 224a_4a_5 \\
& - 56a_5^2 + 18a_7^2 \\
& - 174\Big)
\end{aligned}
$$

$$
f_4 = -2a_0a_3
$$

$$
\begin{aligned}
w_4 = {} & \frac{3}{128}\Big(2a_3a_6 - a_4a_7 \\
& + a_5a_7 - 4a_6a_7\Big)
\end{aligned}
$$

$$
\begin{aligned}
f_7 = {} & -\frac{1}{3}\Big(3a_0a_5 + 18a_1a_4 \\
& + 6a_1a_5 + 6a_2a_4 \\
& + 6a_2a_5 - 10\Big)
\end{aligned}
$$

$$
\begin{aligned}
w_7 = {} & \frac{3}{128}\Big(21a_0a_6 - 72a_1a_6 \\
& - 3a_3a_7 - 64a_4a_6 \\
& - 32a_5a_6\Big)
\end{aligned}
$$

$$
f_9 = 2a_7\left(a_1 - a_2\right)
$$

$$
\begin{aligned}
w_9 = {} & -\frac{3}{128}\Big(a_3a_4 - a_3a_5 \\
& - 6a_6a_7\Big)
\end{aligned}
$$

$$
\begin{aligned}
f_{14} = {} & -48a_1^2 - 48a_1a_2 \\
& - 12a_2^2 + 4a_4a_6 \\
& - 6a_4a_8 + 2a_5a_6 \\
& - 3a_5a_8 - 1
\end{aligned}
$$

$$
\begin{aligned}
w_{14} = {} & -\frac{1}{64}\Big(3a_0^2 - 288a_0a_1 \\
& - 108a_0a_2 - 135a_0a_4 \\
& - 36a_0a_5 - 174a_1^2 \\
& - 516a_1a_2 + 432a_1a_4 \\
& + 108a_1a_5 - 174a_2^2 \\
& - 110\Big)
\end{aligned}
$$

$$
\begin{aligned}
f_{15} = &-16a_1^2 - 16a_1a_2 \\
&- 4a_2^2 - 2a_4a_8 \\
&- a_5a_8 - 1 \\
w_{15} = &\frac{9}{64}\Big(a_0^2 - 96a_0a_1 \\
&- 36a_0a_2 - 21a_0a_4 \\
&- 58a_1^2 - 172a_1a_2 \\
&+ 80a_1a_4 + 4a_1a_5 \\
&- 58a_2^2 - 26\Big) \\
f_{17} = &-a_0^2 + 6a_4a_6 \\
&- 8a_4a_8 + 2a_5a_6 \\
&- 4a_5a_8 + 4 \\
w_{17} = &-\frac{3}{64}\Big(18a_0a_4 + 9a_0a_5 \\
&- 48a_1a_4 - 24a_1a_5 \\
&- 20\Big) \\
f_{18} = &-\frac{1}{2}\Big(2a_0^2 - 2a_3^2 \\
&- a_4a_6 + a_5a_6 \\
&- 4\Big) \\
w_{18} = &-\frac{3}{16} \\
f_{19} = &-a_6\left(a_3 + a_7\right) \\
w_{19} = &\frac{3}{32}\Big(3a_0a_7 - a_1a_3 \\
&+ a_2a_3\Big) \\
f_{25} = &-\frac{1}{2}\Big(2a_1^2 - 4a_1a_2 \\
&+ 2a_2^2 - a_4a_6 \\
&+ a_5a_6 - 2a_7^2 \\
&+ 4\Big) \\
w_{25} = &-\frac{9}{16}
\end{aligned}
$$

With all omitted multipliers equal to zero, direct expansion gives

$$
\boxed{\sum_{i\in\{0,1,2,4,7,9,14,15,17,18,19,25\}} w_i f_i = 1.}
$$

The multiplier degrees are at most two. Thus the residual order-two ideal is the unit ideal over $\mathbb{Q}[a_0, \ldots, a_8]$.

D Claim-status and verification index

The full cross-paper claim ledger is supplement material. Each extracted article will carry a shorter claim table containing only its own theorem dependencies and adjacent fail-closed boundaries.

The residual computation used in the first draft was frozen at repository commit

`e928f257c25099eb534eb34109dfc1dc6a3127a1`.

The global BGG bridge was added in the subsequent theorem revision based at `c471b99f5e3708e692b1c25238f6272c9e29b48f`.

The selected algebraic BV–BFV suspension, polarization, and pairing transfer were frozen at

`8a7e7821f8cd4af5798fd1cb7a962f1da69cdf86`.

The companion snapshot records all three layers, the environment, hashes, expected-failure guards, and command lines. The exhaustive publication verification is run with

```
python3 symbolic/verify_conformal_paper_free.py --reproduce --timeout 1800
```

The claim/evidence index for the terminal covariant theorem is the generated proof ledger at

`covariant_completion/generated/`
`covariant_H4_proof_ledger.md`.

It binds each required claim to its authoritative certificate digest and reproduction command.

Claim	Status	Evidence / boundary
Formal pure-Weyl detour identities	Exact	Branson–Gover theorem plus action-normalized factorization and finite scalar-block audits.
Global smooth deformation cohomology and curvature isomorphism	Exact smooth theorem	The flat BGG fine resolution and $H^\bullet(\mathbb{R}\times S^3)$ give $H^1_{\rm def}=H^2_{\rm def}=0$ and hence (23). This smooth theorem does not itself establish the later energy-mode or field Cauchy completions.
Euclidean polynomial detour ranks, degrees 2–6	Exact finite certificate	Rational matrices verify kernels, ranks, and the 15 residual zero modes; they are an independent convention audit rather than the source of the global theorem.
Algebraic cylinder realization and stable generator coefficients	Exact symbolic theorem	Full coordinate Weyl/Bach calculation at symbolic n, nonzero $E/A/L$ pivots, explicit same-block metric preimages, character exhaustion, invariant form, and cutoff-stable conformal brackets.
Quadratic minimal field-BV/raw-chain identification	Exact centered-buffer certificate	The master-action tangent differential is constructed in full metric and antifield variables. Explicit inverse maps verify $FQ_{\rm BV}=Q_{\rm raw}F$ and $GQ_{\rm raw}=Q_{\rm BV}G$ on all 2170 basis vectors at energies two through five; gauge fixing is a separate proposition and the dual global zero-mode sector is not included here.
Gauge-fixed/nonminimal equivalence	Exact centered-buffer certificate	The stationary conformal-Landau fermion generates an invertible canonical shear. All vector and scalar antighosts, multipliers, and antifield duals are retained; the transported $p_{\rm gf}, \jmath_{\rm gf}, s_{\rm gf}$ contract them exactly. The low-mode certificate verifies $P_Z Q_{\rm gf}=Q_{\rm gf}P_Z$ and preserves all fifteen CKVs.
Canonical bulk endpoint duality and Taub codomain	Exact algebraic theorem	Perfect endpoint pairings give $\operatorname{coker} K^\sharp \cong (\ker K)^*$ with dual compact type $4_{+1}\oplus 7_0\oplus 4_{-1}$. Exact matrices construct Θ, its quotient section, and the direct-sum endpoint decomposition. The quadratic Kuranishi class evaluates to the certified Taub/moment map. The time-slice degree suspension is supplied by the next theorem, not by endpoint duality alone.
Selected algebraic BV–BFV suspension and polarization	Exact algebraic theorem	Canonical BFV compatibility fixes $\tau=\Theta$ and $\lambda=+1$. The positive-frequency Lagrangian ledger gives $\operatorname{Sym}(\mathcal{W}_+\oplus\mathcal{W}_-)\otimes\Lambda^\bullet\mathfrak{g}^*$, represents b_a by ι_a, and contracts every local and nonminimal doublet to its vacuum. One matter normalization and the cotangent orientation transfer the pairing and give $G_{\rm field}=I_2$.
Completed residual Cartan homotopy reduction	Exact energy-mode theorem	The maximal block-direct-sum $\overline{Q}$ is closed and nilpotent, every total-degree block is finite, and bounded ι_D/δ gives closed range and unchanged centered cohomology. This is a quotient only in the full residual-gauging problem.

Claim	Status	Evidence / boundary
Centered vacuum and one-particle sectors	Exact	Semisimple Lie-algebra cohomology for the vacuum; cutoff-complete absolute rank certificate for one particle.
Centered two-particle sector and $G_{\mathrm{res}} = I_2$	Exact residual theorem	Independent relative and absolute calculations; empty incoming absolute space and exact rank 53.
Vertex descent and dynamical/topological split	Exact locally	Weight-four descent, Chern–Weil transgression, and Euler–Lagrange quotient. Global theta effects are retained.
Energy-mode one-particle/Fock Krein completion	Exact	The normalized $E/A/L$ basis gives an infinite-index Krein space, $\Gamma_s(\mathfrak{J}_1)$ gives the bosonic Fock fundamental symmetry, and weighted-shift domains give closed Krein-adjoint conformal generators on a common core.
Reduced Lorentzian Cauchy–Sobolev realization	Exact cylinder theorem	Tensor/vector curl factorization, action-derived residues, exact $E/A/L$ multiplicities, and normalized harmonics give (54) and a Krein-unitary map to the energy-mode one-particle completion. The physical TT and vector blocks are Green hyperbolic.
Auxiliary 66-to-30 Fourier SDR	Exact flat/Fourier theorem	The exact ghost biwave, flat scalar symbols, support-local Stueckelberg/auxiliary reduction, and symbol-level formal adjoints are certified. This row does not assert a curved scalar-wave symbol.
Curved auxiliary operator, retract, and current identities	Exact cylinder theorems	The action-derived curved Hessian and four-row identities, the all-row BV-canonical support-local SDR, and the $d + Q$ current comparison are exact. Thus `curved_operator_identity`, `curved_deformation_retract`, and `curved_current_comparison` are true.
Scalar-wave witness on the 24-field auxiliary bundle	Exact no-go theorem	At a generic null covector rank $E_2 = 11 > 9 = \operatorname{rank} K_1$; hence no invertible pointwise J and first-order C can make $J^{-1}E_2 + K_1C_1$ vanish on the null cone.
Physical null-symbol quotient and Weyl carrier	Exact symbol theorem	The quotient has dimension two and transverse helicities $+2$ and -2. The reduced Weyl symbol is $\frac{1}{4}I_2$. No unreduced claim $\ker \mathcal{W}_2 = \operatorname{im} K_1$ is made.
Electric/magnetic Weyl–Cotton first-order system	Exact curved theorem	The Cotton divergence supplies the required sixteen components. The exact 34×26 covariant table agrees with all 150 Weyl two-jets. Its eight primary and six secondary constraints give a formally equivalent adjusted 26-state system with positive symmetrizer, causal speeds, exact sourced compatibility, and constraint propagation. The all-level $E/A/L$ spectrum and absence of Cotton duplication are symbolic theorems.

Claim	Status	Evidence / boundary
Prenormal auxiliary symbol and relative saddle diagnostics	Exact fail-closed diagnostics	The generic principal symbol obeys $(P_2 - qI)^2 = 0$ with algebraic/wave/biwave Smith multiplicities 6/12/6, but the naive frozen lower completion has nonzero lower-order remainder. The complete invariant ansatz has $\dim D_0 = 38$ and $\dim D_1 = 93$; simultaneous DP/PD cubic divisibility leaves a 45-parameter family and hence no cubic obstruction. The complete 421-variable sparse quadratic PBW system is assembled; its order-two 2484×190 algebraic matrix has rank 100, and exact cokernel projection yields 2130 nonzero polynomial constraints with no constant obstruction. The projected and lower-order systems remain open. The corrected spatial-projector curvature sign passes focused primal coordinate-jet tests, and exact symmetrized-jet PBW inversion certifies the composition backend. The 48-entry naive sorted-table defect is resolved by the pairing-aware symmetrized-PBW adjoint: imposing $D^\sharp = D$ leaves 44/24 first/zeroth-order complement dimensions, a 21-dimensional cubic family and a 214-parameter nonlinear sharp branch. Its exact order-two Schur gate has rank 52/100 and 1050 nonzero projected constraints, but no constant obstruction or affine-linear consequence. The four bare-$\square$ orientations are exactly inconsistent with rank 159/160 and a common one-row $f_{01} \to f_{00}$ obstruction. A two-direction bilinear split repairs its full $SO(3)$ orbit, but that fixed minimal split fails the complete order-two system with rank 100/101 and a one-row $h_{22} \to f_{01}$ obstruction. This does not decide the general two-nontrivial-factor branch. The smallest reciprocal relative saddle is the odd-adjoint pair $4 + 5$. Its Schur complement is nonlocal. The natural $A_F = p_F A_{\mathrm{eq}}$, $S = A_F^\sharp$, $R = A_F^\sharp J_U$ realization has balanced timelike rank at most 107/116, zero temporal leading coefficient and no positive symmetrizer. This excludes only that realization, not larger relative witnesses or a support-local first-order prolongation. The complete 162-dimensional expanded-relative family contains three reciprocal two-pair candidates, but its central cross-ranks are only 21/24; the three missing directions are rotation scalars which must be retained or locally prolonged.

Claim	Status	Evidence / boundary
Curvature causal homotopy and final covariant H^4	Exact covariant theorem; two legacy implementation flags false	Thirteen flags in (103) are true. The local all-row prolongation, BV differential, off-shell current comparison, and 356/30 hybrid algebraic projector are complete. The retained $5 \to 10 \to 10 \to 5$ endpoint is coefficient-complete and cyclic. Its corrected canonical backward witness has trace-free scalar biwave symbol. The complete parallel invariant two-wave-factor family is excluded by a twelve-multiplier degree-two Nullstellensatz certificate. Its upper curvature graph lift is exact; the canonical middle lift is obstructed. The minimal cyclic relative saddle exists on all 386 rows but has zero endpoint Schur correction, so it neither helps nor obstructs the still-open endpoint-diagonal Green problem. The curved adjoint-tractor HPL compression resolves the former 48 entries in ten invariant channels and matches the full Bach block at orders zero through four with normalization -2. Parent transfer and the explicit trace/Weyl shear give an exact causal homotopy on all 386 rows. No canonical D_{TF} Green inverse or prolonged witness is claimed or needed for that identity. Its causal difference is a compact-to-global quasi-isomorphism; cutoff sources recover the fifteen CKVs and dual endpoints with no duplicate copy, and the cutoff commutator gives $SO(4,2)$ transport up to chain homotopy. The direct cyclic Green/current certificate identifies the causal, prolonged, auxiliary, metric and $E/A/L$ pairings. Consequently `final_covariant_H4` is true and transports the already-complete residual $H^4 = \mathbb{C}^2$ and $G = I_2$ without recomputation. The false `prolonged_green_witness` and `curvature_causal_green_operators` flags are scoped to the superseded canonical implementation; no Hadamard theorem, group exponentiation, or uniqueness among boundary, clock, asymptotic, or deparametrized polarizations is established.
Interpretation as a physical state space	Choice-dependent	Requires a boundary and gauge principle that treats all fifteen generators, including D, as gauge rather than global charges.
Quantum anomaly freedom and interacting probability	Not claimed	Outside the scope of this paper.

References

[1] T. Branson and A. R. Gover, "The conformal deformation detour complex for the obstruction tensor," arXiv:math/0605192.

[2] A. R. Gover and L. J. Peterson, "The ambient obstruction tensor and the conformal deformation complex," Pacific J. Math. **226** (2006) 309–351, arXiv:math/0408229.

[3] A. R. Gover, P. Somberg, and V. Souček, "Yang–Mills detour complexes and conformal geometry," Commun. Math. Phys. **278** (2008) 307–327, arXiv:math/0606401.

[4] M. Benini, G. Musante, and A. Schenkel, "Green hyperbolic complexes on Lorentzian manifolds," Commun. Math. Phys. (2023), doi:10.1007/s00220-023-04807-5, arXiv:2207.04069.

[5] C. Bär, "Green-hyperbolic operators on globally hyperbolic spacetimes," Commun. Math. Phys. **333** (2015) 1585–1615, arXiv:1310.0738.

[6] P. E. T. Jorgensen and F. Tian, "Unbounded operators, Lie algebras, and local representations," J. Math. Phys. **55** (2014) 111702, arXiv:1406.6966.

[7] A. Čap, "Overdetermined systems, conformal geometry, and the BGG complex," in *Symmetries and Overdetermined Systems of Partial Differential Equations*, IMA Vol. Math. Appl. **144** (Springer, 2008) 1–24, arXiv:math/0610225.

[8] A. Čap, "Infinitesimal automorphisms and deformations of parabolic geometries," J. Eur. Math. Soc. **10** (2008) 415–437, arXiv:math/0508535.

[9] A. Čap and K. Hu, "Bounded Poincaré operators for twisted and BGG complexes," J. Math. Pures Appl. **179** (2023) 253–276, arXiv:2304.07185.

[10] K.-j. Hamada and S. Horata, "Conformal algebra and physical states in non-critical 3-brane on $\mathbb{R} \times S^3$," Prog. Theor. Phys. **110** (2003) 1169–1210, arXiv:hep-th/0307008.

[11] K.-j. Hamada, "Building blocks of physical states in a non-critical 3-brane on $\mathbb{R} \times S^3$," Int. J. Mod. Phys. A **20** (2005) 5353–5398, arXiv:hep-th/0402136.

[12] K.-j. Hamada, "Conformal field theory on $\mathbb{R} \times S^3$ from quantized gravity," Int. J. Mod. Phys. A **24** (2009) 3073–3110, arXiv:0811.1647.

[13] K.-j. Hamada, "BRST analysis of physical fields and states for 4D quantum gravity on $\mathbb{R} \times S^3$," Phys. Rev. D **86** (2012) 124006, arXiv:1202.4538.

[14] S. M. Kuzenko and M. Ponds, "Conformal geometry and (super)conformal higher-spin gauge theories," JHEP **05** (2019) 113, arXiv:1902.08010.

[15] M. Beccaria, X. Bekaert, and A. A. Tseytlin, "Partition function of free conformal higher spin theory," JHEP **08** (2014) 113, arXiv:1406.3542.

[16] R. R. Metsaev, "Ordinary-derivative formulation of conformal low-spin fields," JHEP **01** (2008) 064, arXiv:0707.4437.

[17] T. Nutma and M. Taronna, "On conformal higher spin wave operators," JHEP **06** (2014) 066, arXiv:1404.7452.

[18] N. Boulanger and M. Henneaux, "A derivation of Weyl gravity," Annalen Phys. **10** (2001) 935–964, arXiv:hep-th/0106065.

[19] G. Barnich, F. Brandt, and M. Henneaux, "Local BRST cohomology in gauge theories," Phys. Rept. **338** (2000) 439–569, arXiv:hep-th/0002245.

[20] G. Barnich, F. Brandt, and M. Henneaux, "Local BRST cohomology in the antifield formalism: I. General theorems," Commun. Math. Phys. **174** (1995) 57–92, arXiv:hep-th/9405109.

[21] G. Barnich, M. Henneaux, T. Hurth, and K. Skenderis, "Cohomological analysis of gauged-fixed gauge theories," Phys. Lett. B **492** (2000) 376–384, arXiv:hep-th/9910201.

[22] Y. Décanini and A. Folacci, "FKWC-bases and geometrical identities for classical and quantum field theories in curved spacetime," arXiv:0805.1595.

[23] J. M. Martín-García, D. Yllanes, and R. Portugal, "The Invar tensor package: Differential invariants of Riemann," arXiv:0802.1274.

[24] A. S. Cattaneo, P. Mnev, and N. Reshetikhin, "Classical BV theories on manifolds with boundary," Commun. Math. Phys. **332** (2014) 535–603, arXiv:1201.0290.

[25] V. Iyer and R. M. Wald, "Some properties of Noether charge and a proposal for dynamical black hole entropy," Phys. Rev. D **50** (1994) 846–864, arXiv:gr-qc/9403028.

[26] R. R. Metsaev, "BRST–BV approach to conformal fields," J. Phys. A **49** (2016) 175401, arXiv:1511.01836.

[27] I. Dneprov and M. Grigoriev, "Presymplectic BV–AKSZ formulation of conformal gravity," doi:10.1140/epjc/s10052-022-11082-6, arXiv:2208.02933.

[28] G. Barnich, X. Bekaert, and M. Grigoriev, "Notes on conformal invariance of gauge fields," J. Phys. A **48** (2015) 505402, arXiv:1506.00595.

[29] E. Altas and B. Tekin, "Linearization instability for generic gravity in AdS," Phys. Rev. D **97** (2018) 024028, arXiv:1705.10234.

[30] E. Altas and B. Tekin, "Second order perturbation theory in general relativity: Taub charges as integral constraints," Phys. Rev. D **99** (2019) 104078, arXiv:1903.11982.

[31] C. Chevalley and S. Eilenberg, "Cohomology theory of Lie groups and Lie algebras," Trans. Amer. Math. Soc. **63** (1948) 85–124, doi:10.1090/s0002-9947-1948-0024908-8.

[32] W. Fischler and S. Kundu, "Physical effects of the gravitational Θ-parameter," JHEP **05** (2017) 098, arXiv:1612.06010.